

# Forecaster's Toolbox

03

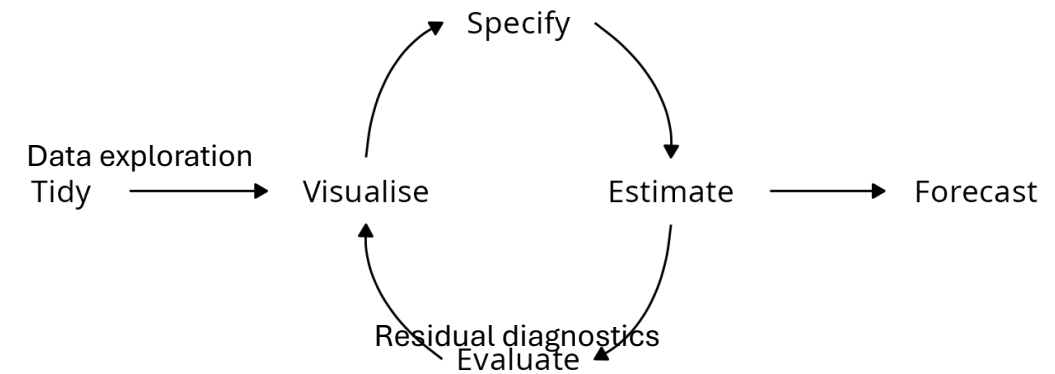
4 Time series features

5 The forecaster's toolbox

# Outline

- Data exploration & preparation
  - (20) Lagged variables
- Specify
  - (30) Autocorrelation function (ACF)
  - (15) White noise
- (35) Evaluate: Time series cross validation
- (20) Evaluate: Model assumptions
  - Random error terms & Residual diagnostics

## Forecasting workflow



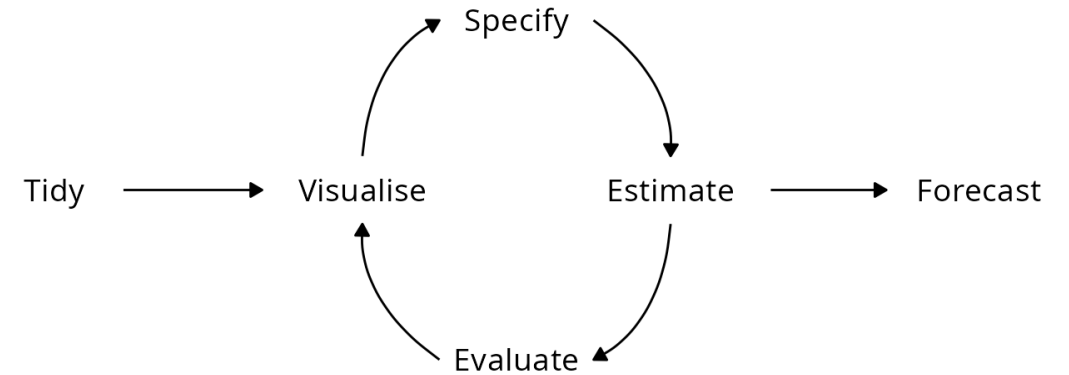
# Data exploration & preparation

Lagged variables

# Forecasting workflow

## Data preparation (tidy)

- Transformations and adjustments (last lecture)
- Differencing
- Lagged variables
- Missing data
  - Imputation
  - Linear interpolation
- Outlier



### Visualise

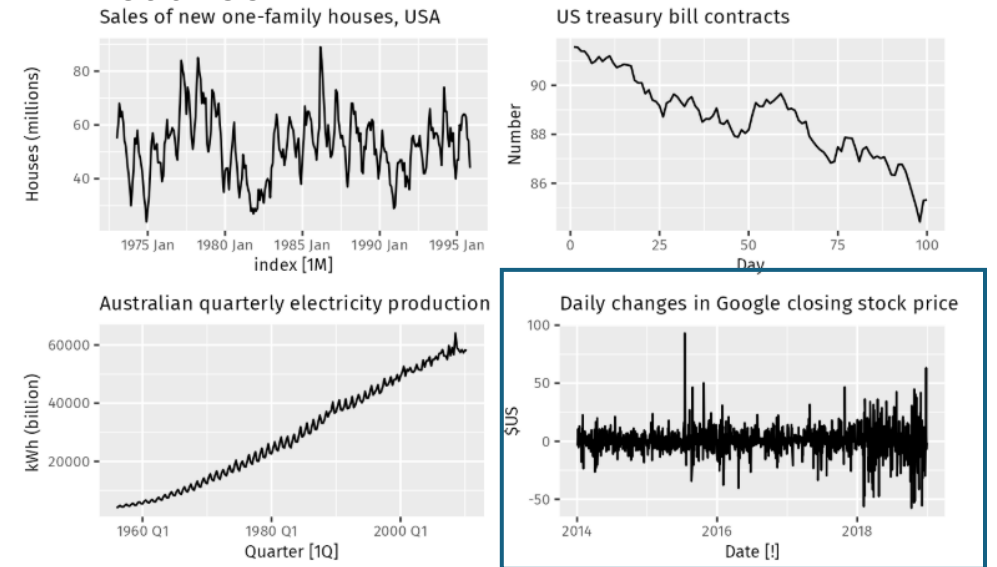
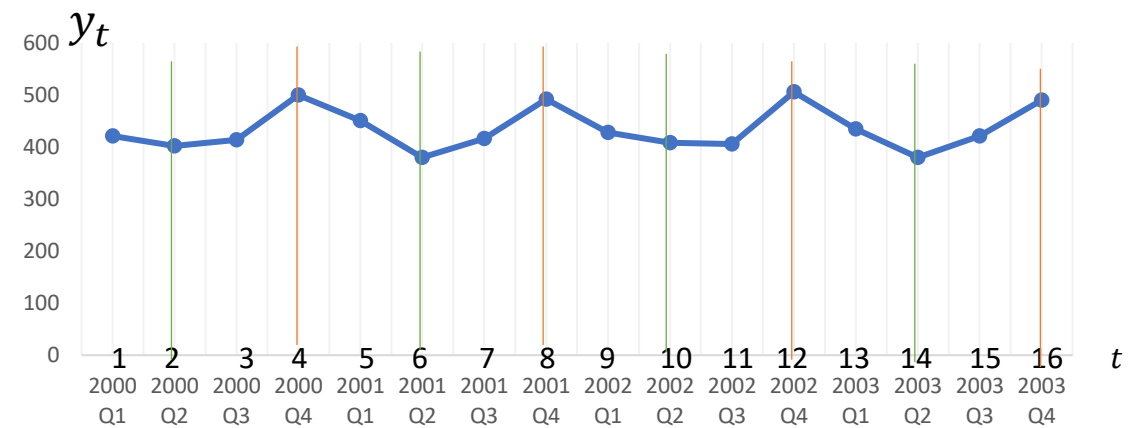


Figure 2.3: Four examples of time series showing different patterns.

# Lagged values of time series

File: 04-FCToolbox

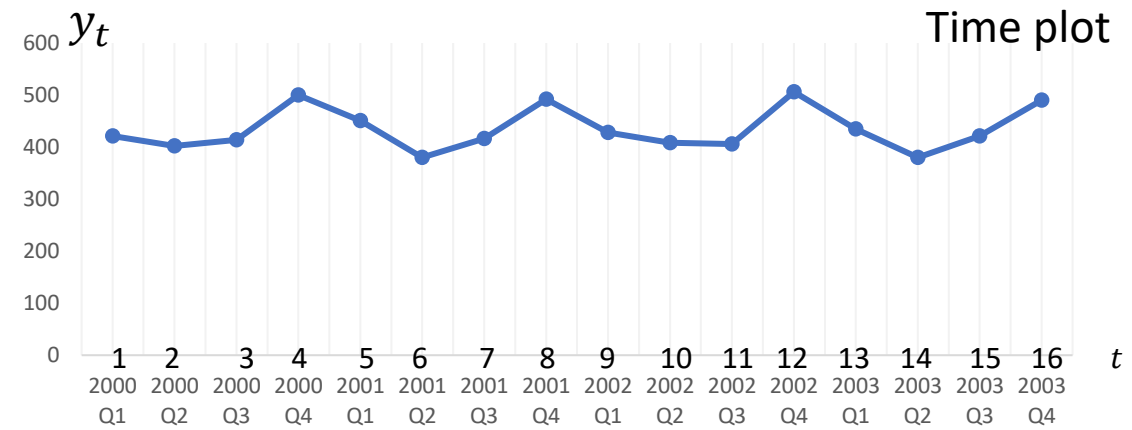
Sheet: Lag



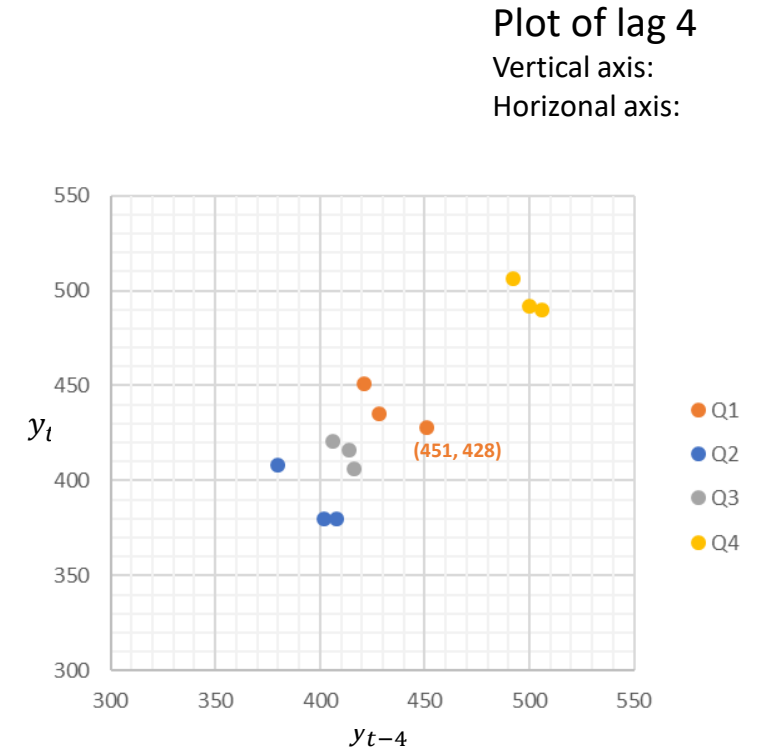
| Quarter K | $t$ | Observation<br>$y_t$ | Lag1<br>$By_t = y_{t-1}$ | Lag2<br>$B^2y_t = y_{t-2}$ | Lag3<br>$B^3y_t = y_{t-3}$ | Lag4 |
|-----------|-----|----------------------|--------------------------|----------------------------|----------------------------|------|
| 2000 Q1   | 1   | 421                  |                          |                            |                            |      |
| 2000 Q2   | 2   | 402                  | 421                      |                            |                            |      |
| 2000 Q3   | 3   | 414                  | 402                      | 421                        |                            |      |
| 2000 Q4   | 4   | 500                  | 414                      | 402                        | 421                        |      |
| 2001 Q1   | 5   | 451                  | 500                      | 414                        | 402                        | 421  |
| 2001 Q2   | 6   | 380                  | 451                      | 500                        | 414                        | 402  |
| 2001 Q3   | 7   | 416                  | 380                      | 451                        | 500                        | 414  |
| 2001 Q4   | 8   | 492                  | 416                      | 380                        | 451                        | 500  |
| 2002 Q1   | 9   | 428                  | 492                      | 416                        | 380                        | 451  |
| 2002 Q2   | 10  | 408                  | 428                      | 492                        | 416                        | 380  |
| 2002 Q3   | 11  | 406                  | 408                      | 428                        | 492                        | 416  |
| 2002 Q4   | 12  | 506                  | 406                      | 408                        | 428                        | 492  |
| 2003 Q1   | 13  | 435                  | 506                      | 406                        | 408                        | 428  |
| 2003 Q2   | 14  | 380                  | 435                      | 506                        | 406                        | 408  |
| 2003 Q3   | 15  | 421                  | 380                      | 435                        | 506                        | 406  |
| 2003 Q4   | 16  | 490                  | 421                      | 380                        | 435                        | 506  |



# Lag plot



| Quarter | $t$ | Observation<br>$y_t$ | Lag1<br>$By_t = y_{t-1}$ | Lag2<br>$B^2y_t = y_{t-2}$ | Lag3 | Lag4 |
|---------|-----|----------------------|--------------------------|----------------------------|------|------|
| 2000 Q1 | 1   | 421                  |                          |                            |      |      |
| 2000 Q2 | 2   | 402                  | 421                      |                            |      |      |
| 2000 Q3 | 3   | 414                  | 402                      | 421                        |      |      |
| 2000 Q4 | 4   | 500                  | 414                      | 402                        | 421  |      |
| 2001 Q1 | 5   | 451                  | 500                      | 414                        | 402  | 421  |
| 2001 Q2 | 6   | 380                  | 451                      | 500                        | 414  | 402  |
| 2001 Q3 | 7   | 416                  | 380                      | 451                        | 500  | 414  |
| 2001 Q4 | 8   | 492                  | 416                      | 380                        | 451  | 500  |
| 2002 Q1 | 9   | 428                  | 492                      | 416                        | 380  | 451  |
| 2002 Q2 | 10  | 408                  | 428                      | 492                        | 416  | 380  |
| 2002 Q3 | 11  | 406                  | 408                      | 428                        | 492  | 416  |
| 2002 Q4 | 12  | 506                  | 406                      | 408                        | 428  | 492  |
| 2003 Q1 | 13  | 435                  | 506                      | 406                        | 408  | 428  |
| 2003 Q2 | 14  | 380                  | 435                      | 506                        | 406  | 408  |
| 2003 Q3 | 15  | 421                  | 380                      | 435                        | 506  | 406  |
| 2003 Q4 | 16  | 490                  | 421                      | 380                        | 435  | 506  |



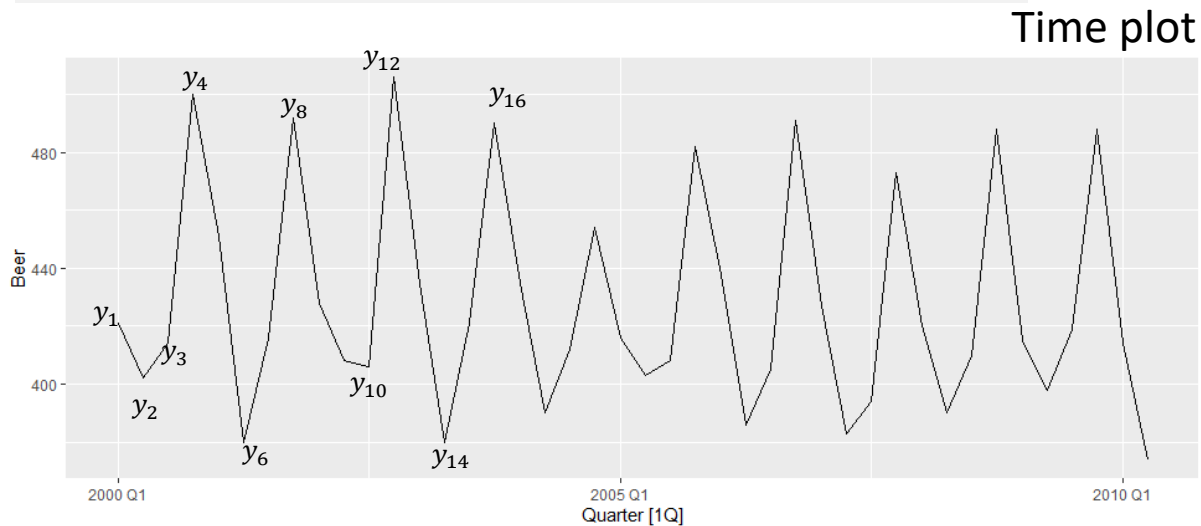
• Plot of lag2

```
glimpse(aus_production)

recent_production <- aus_production %>%
  filter(year(Quarter) >= 2000)

autoplot(recent_production, Beer)

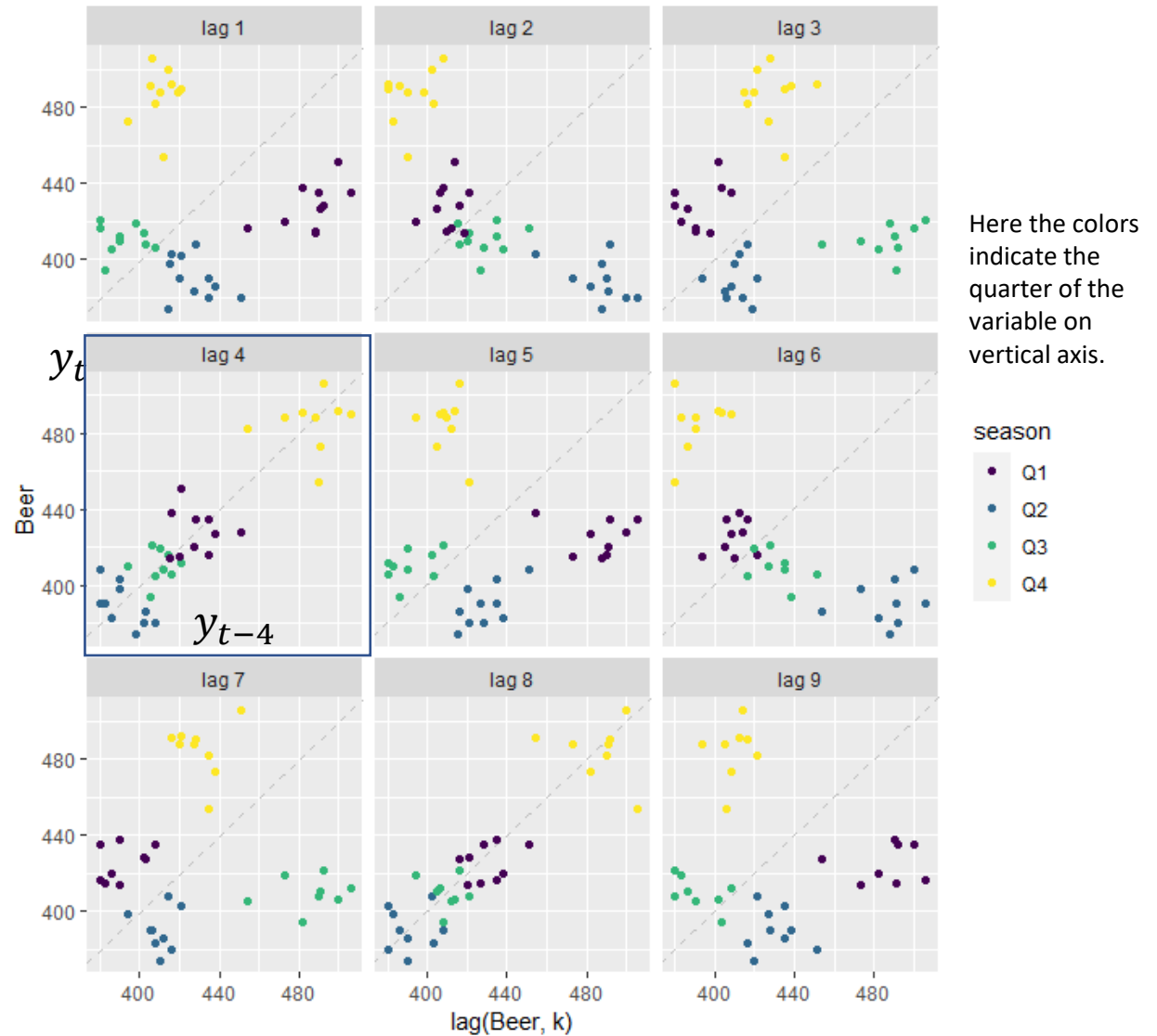
recent_production %>%
  gg_lag(Beer, geom = "point") +
  labs(x = "lag(Beer, k)")
```



Peaks (high points) at Q4, i.e.,  $t = 4, 8, 12, 16, \dots$   
 Troughs (low points) at Q2, i.e.,  $t = 2, 6, 10, 14, \dots$

- Strong positive relationship at lags 4 and 8
- Negative relationship at lags 2 and 6.

# Lag plots

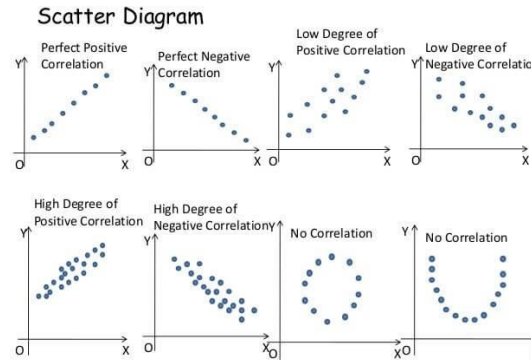


Specify: Autocorrelation function (ACF)



# Autocorrelation: “memory of the series”

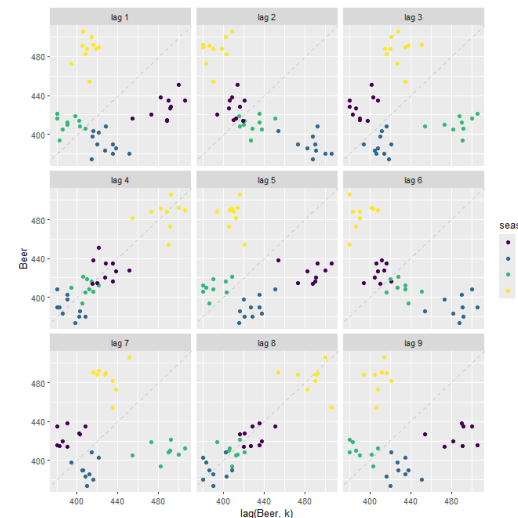
- Recall sample **covariance** between  $x$  and  $y$
- Recall sample **correlation** between  $x$  and  $y$



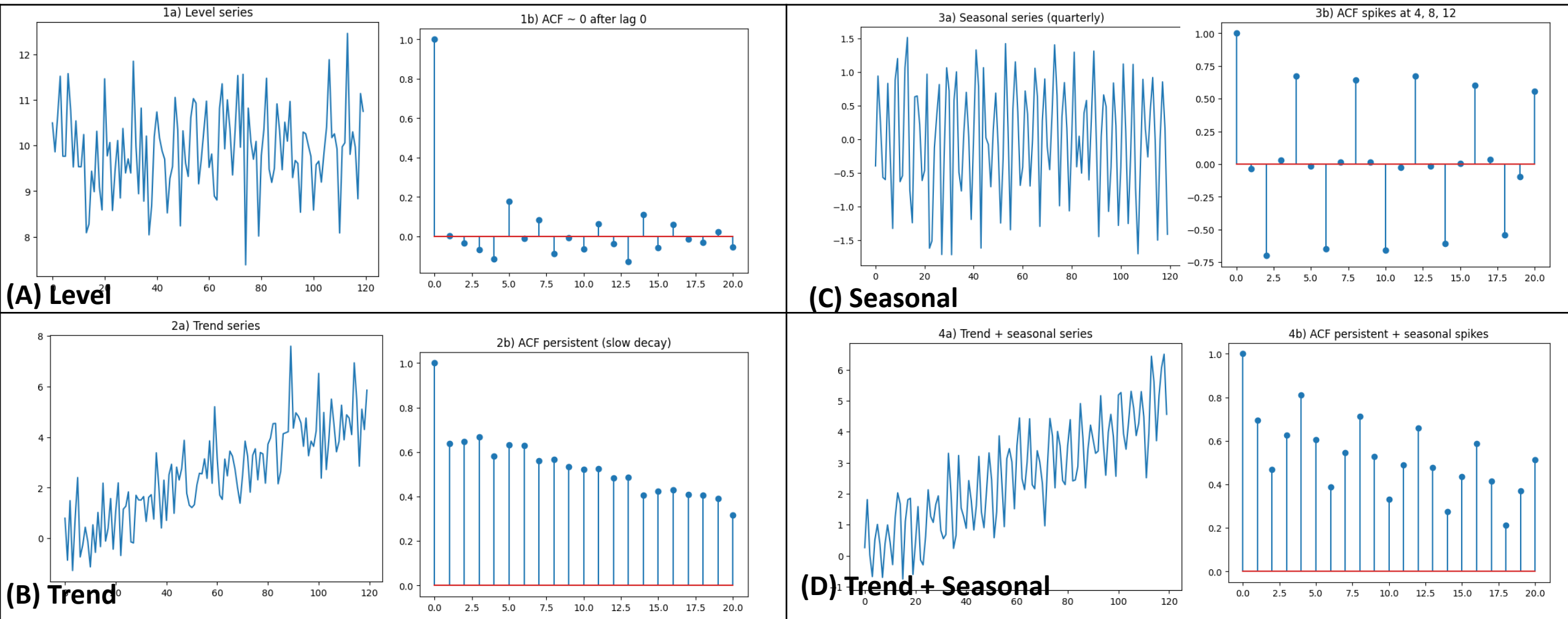
$$cov_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}$$

- Sample **autocorrelation**  $r_k$  is almost the same as the sample correlation between  $y_t$  and  $y_{t-k}$
- Like the correlation coefficient,  $r_k \in [-1, 1]$ .
- If  $r_k = 0$ , no linear relationship between  $y_t$  and  $y_{t-k}$ . No “memory”



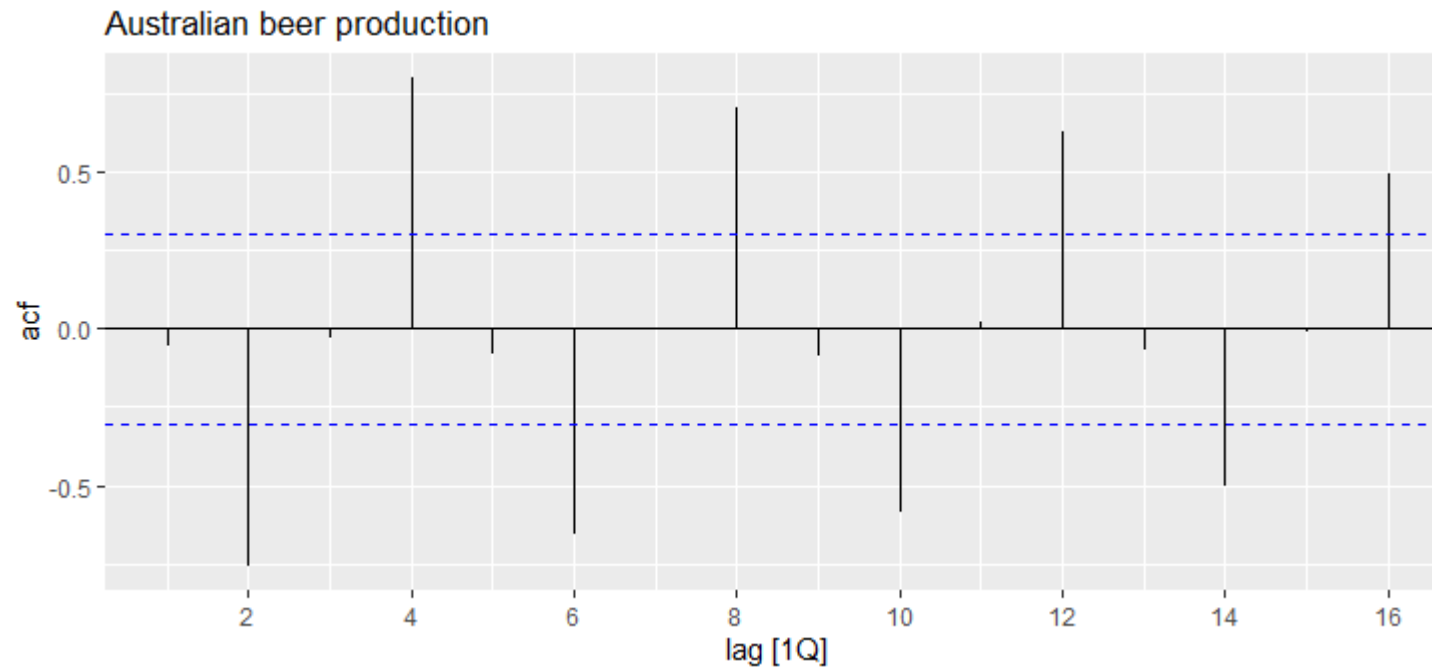
$$r_k = \frac{\sum_{t=k+1}^T (y_t - \bar{y})(y_{t-k} - \bar{y})}{\sum_{t=1}^T (y_t - \bar{y})^2}$$



| Time-series feature      | What you see in ACF          | What it means                  |
|--------------------------|------------------------------|--------------------------------|
| No trend, no seasonality | ACF $\approx 0$ after lag 0  | Memoryless $\rightarrow$ Level |
| Trend                    | Slow decay at small lags     | Persistence                    |
| Seasonality (quarterly)  | Spikes at lags 4, 8, 12      | Seasonal repetition            |
| Trend + seasonality      | Slow decay + seasonal spikes | Both present                   |

# Correlogram autocorrelation function (acf)

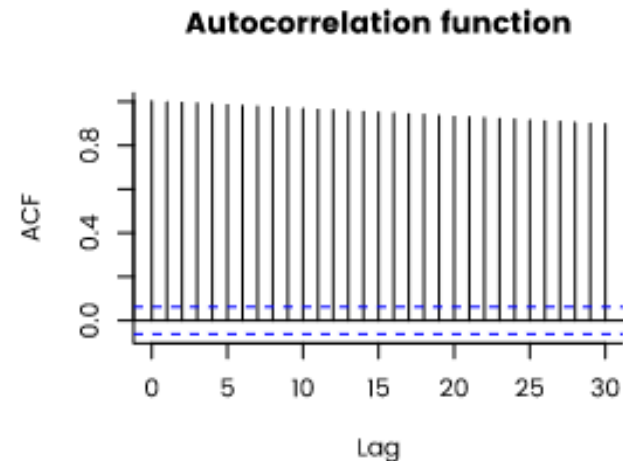
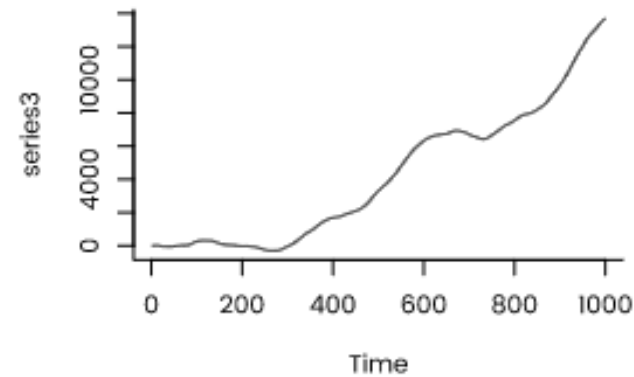
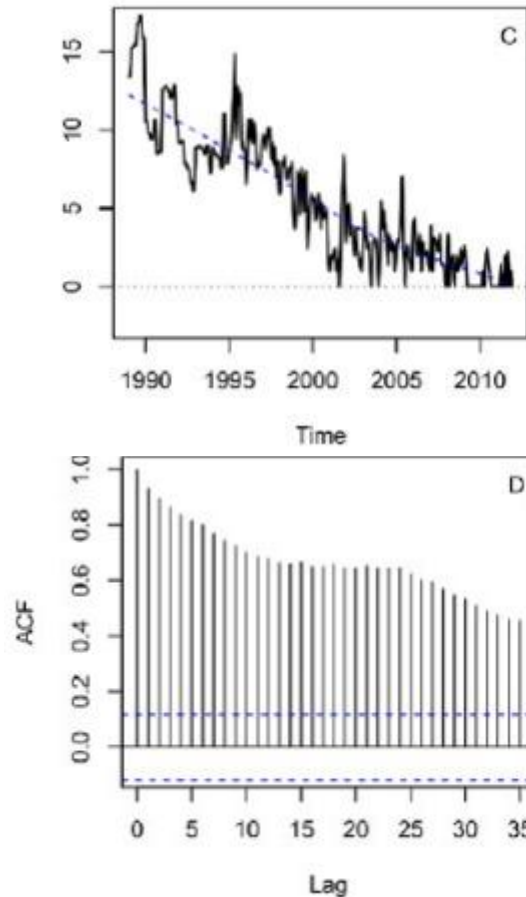
```
recent_production %>%  
  ACF(Beer) %>%  
  autoplot() + labs(title="Australian beer production")
```



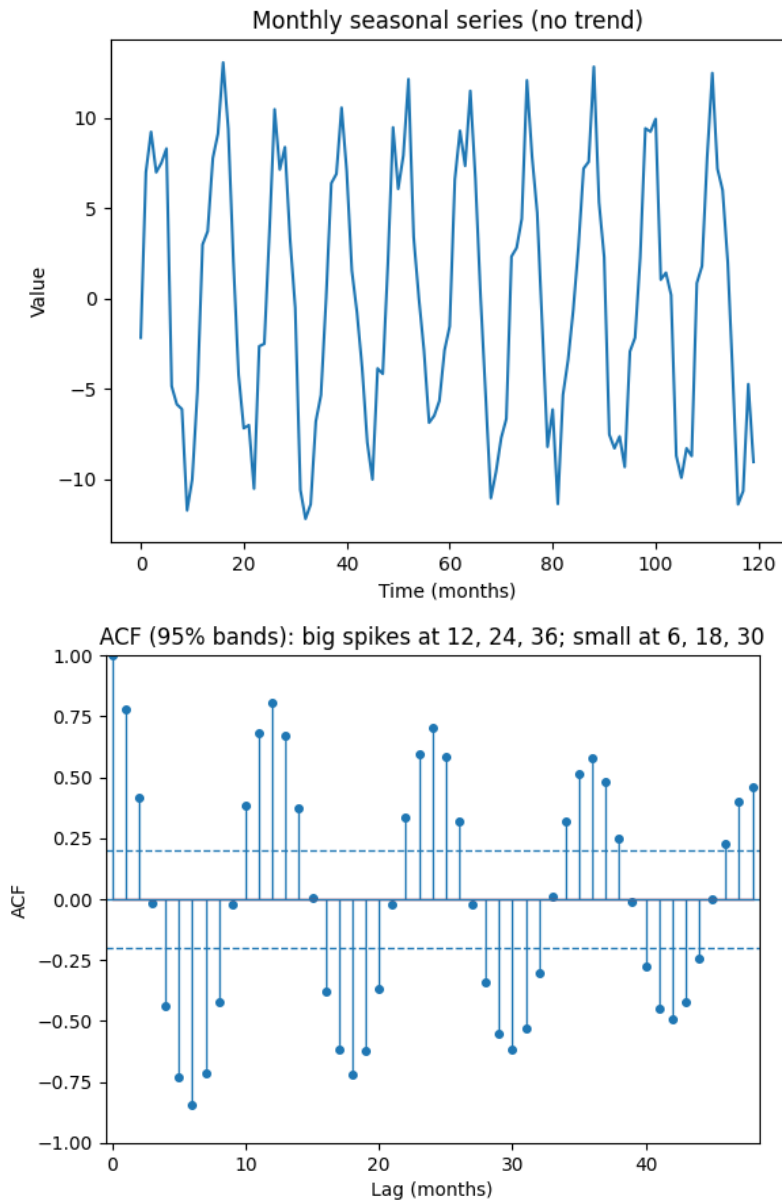
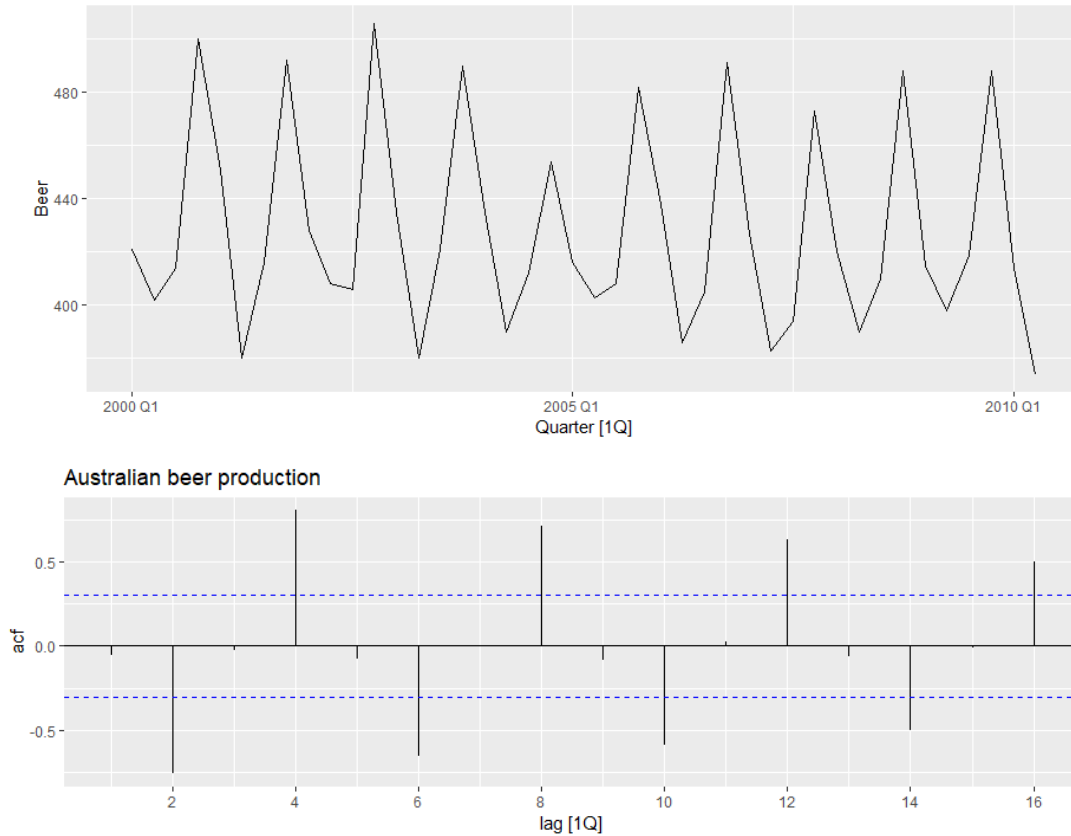
The dashed blue lines indicate whether the correlations are significantly different from zero

# Trend and seasonality in ACF plots

- When data have a trend, the autocorrelations for small lags tend to be large and positive.



# Seasonality in ACF plots



When data are seasonal, the autocorrelations will be larger at the seasonal lags (i.e., at multiples of the seasonal frequency)

# Trend and seasonality in ACF plots

- When data have a trend, the autocorrelations for small lags tend to be large and positive.
- When data are seasonal, the autocorrelations will be larger at the seasonal lags (i.e., at multiples of the seasonal frequency)
- When data are trended and seasonal, you see a combination of these effects.
- Generally, as the number of lags increases, the magnitudes of the autocorrelation coefficients decrease.

```
autoplot(a10, Cost) +  
  labs(y = "$ (millions)",  
       title = "Australian antidiabetic drug sales")
```

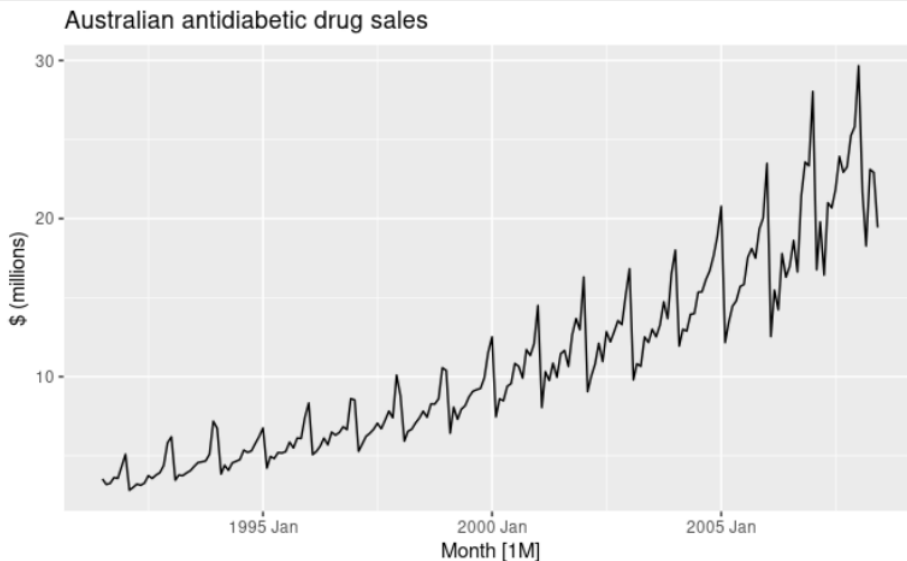
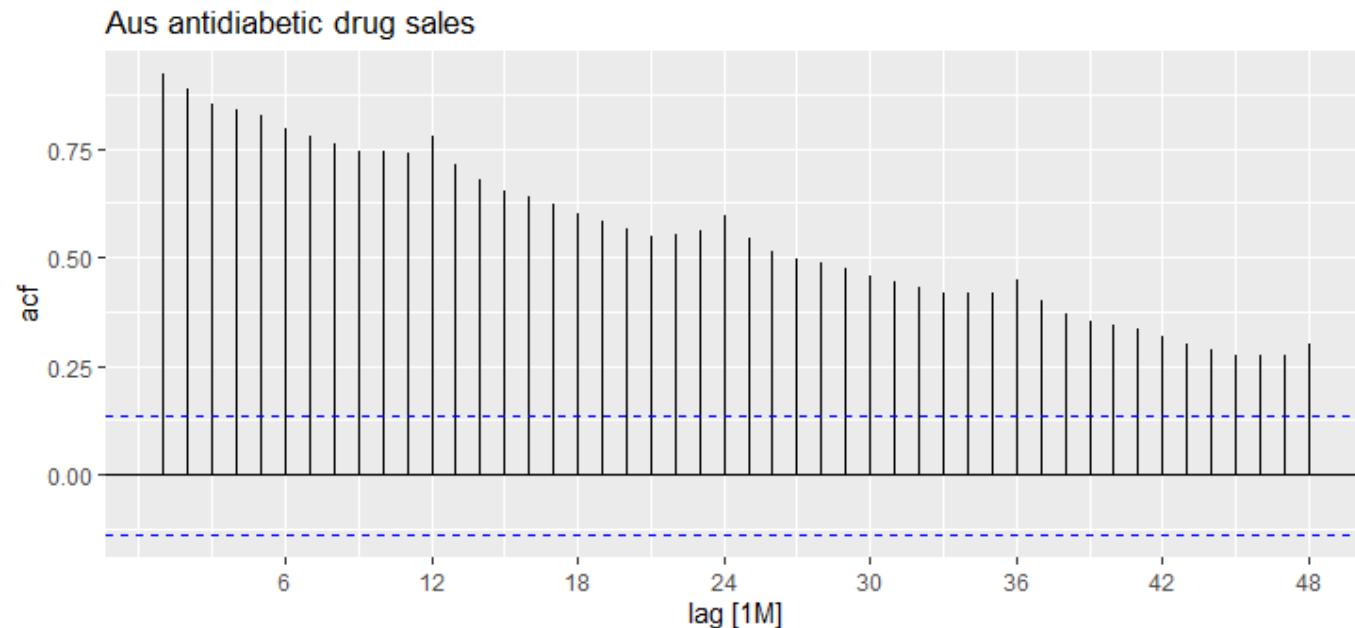


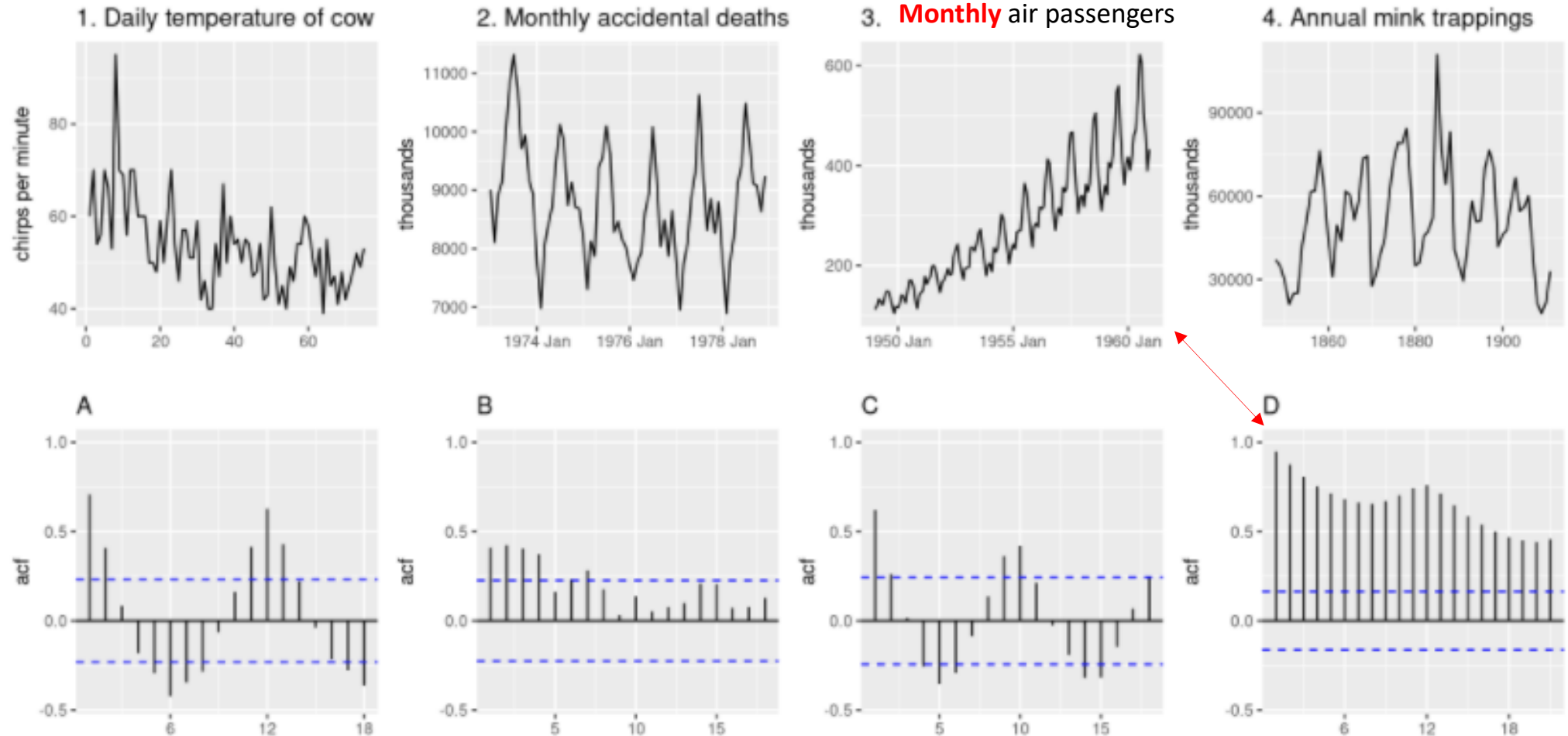
Figure 2.2: Monthly sales of antidiabetic drugs in Australia.

```
a10 %>%  
  ACF(Cost, lag_max = 48) %>%  
  autoplot() +  
  labs(title="Aus antidiabetic drug sales")
```



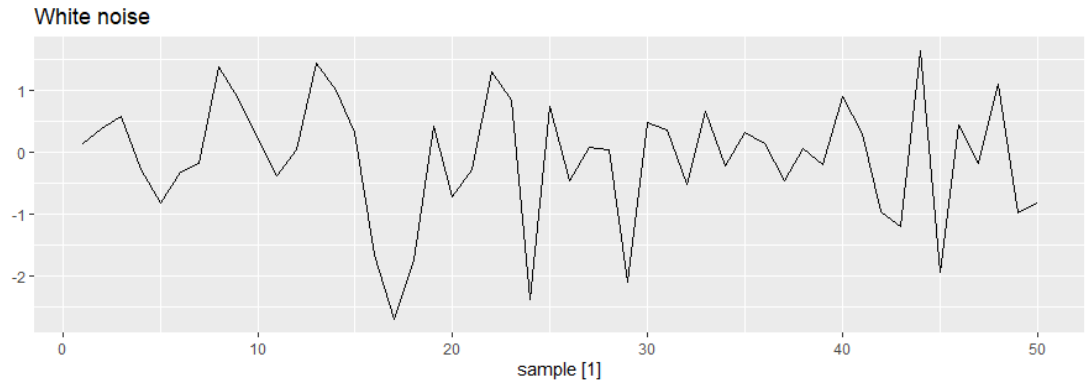
The following time plots and ACF plots correspond to four different time series. Your task is to match each time plot in the first row with one of the ACF plots in the second row.

In-class: 1 \_\_\_, 2 \_\_\_, 3D, 4 \_\_\_



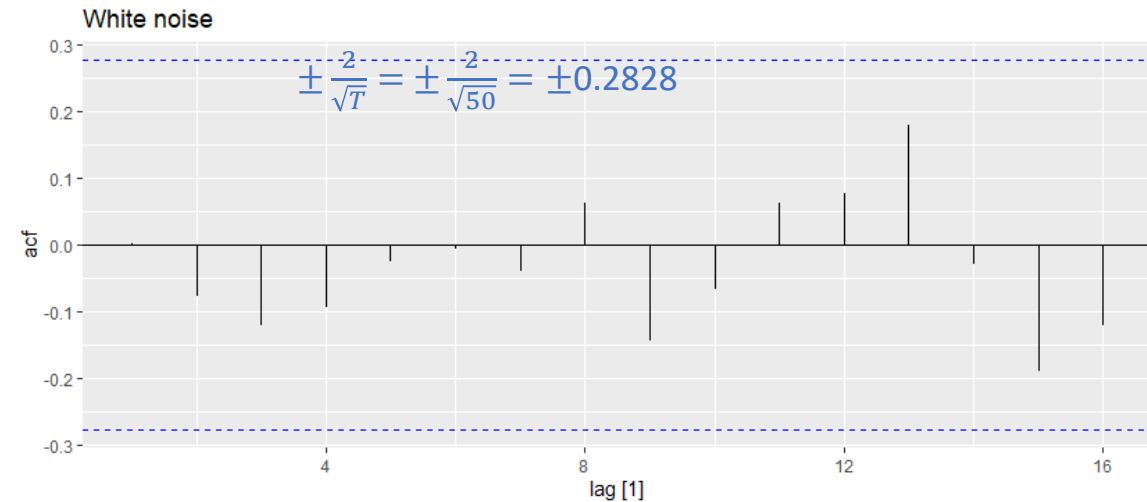
# White *noise* Time series that show no autocorrelation → “no memory”

```
set.seed(37)
y <- tsibble(sample = 1:50, wn = rnorm(50), index = sample)
y %>% autoplot(wn) + labs(title = "White noise", y = "")
```



```
y %>%
  ACF(wn) %>%
  autoplot() + labs(title = "White noise")
```

Here,  $T = 50$ , the critical values are



- For white noise series, we expect 95% of the spikes in the ACF to lie within  $\pm 2/\sqrt{T}$  (called the **critical values**) where  $T$  is the length of time series.
- If one or more large spikes are outside these bounds, or if substantially more than 5% of spikes are outside these bounds, then the series is probably not white noise.

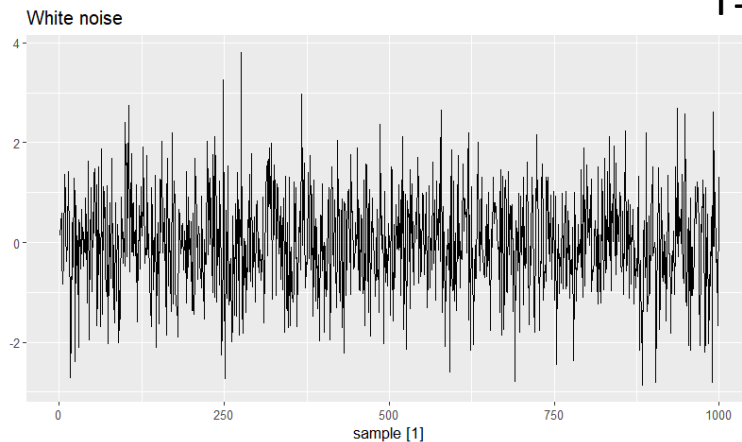
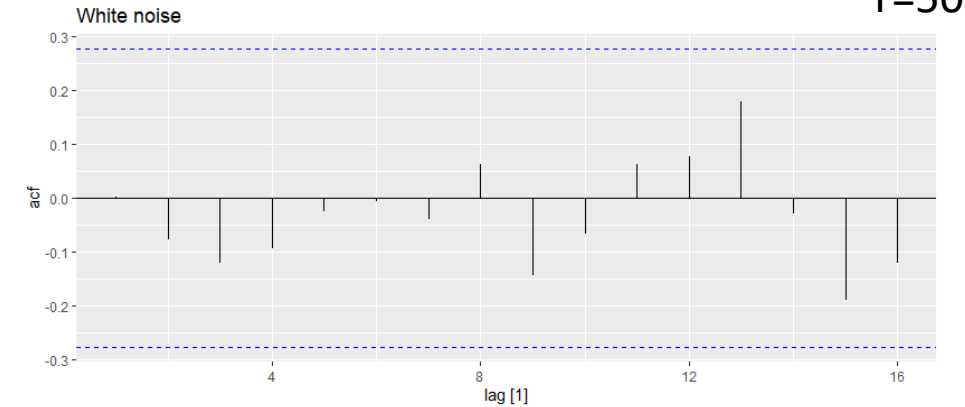
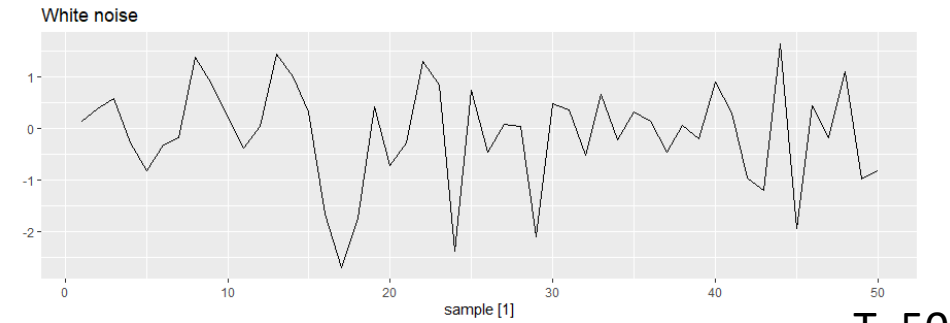


# White noise

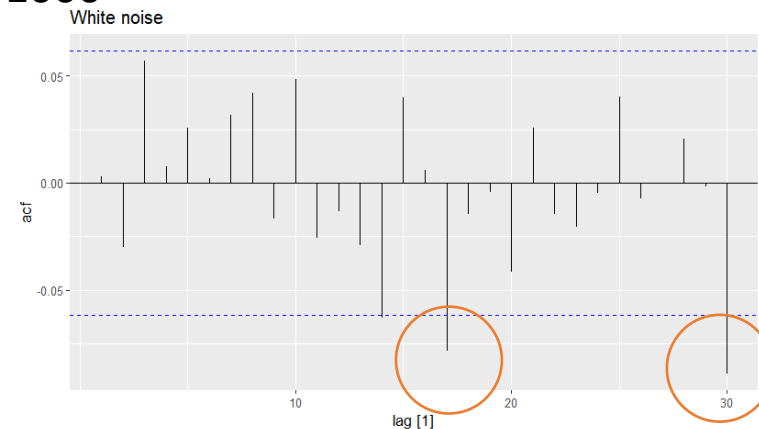
```
set.seed(37)
y <- tsibble(sample = 1:50, wn = rnorm(50), index = sample)
y %>% autoplot(wn) + labs(title = "White noise", y = "")

y %>%
  ACF(wn) %>%
  autoplot() + labs(title = "White noise")
```

- For white noise series, we expect 95% of the spikes in the ACF to lie within  $\pm 2/\sqrt{T}$  (called the **critical values**) where  $T$  is the length of time series.
- If one or more large spikes are outside these bounds, or if substantially more than 5% of spikes are outside these bounds, then the series is probably not white noise.



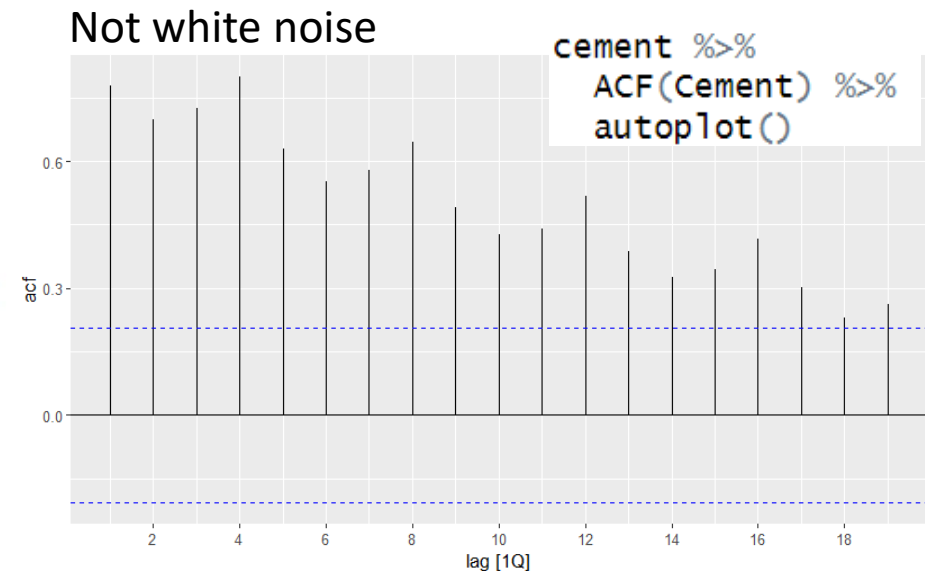
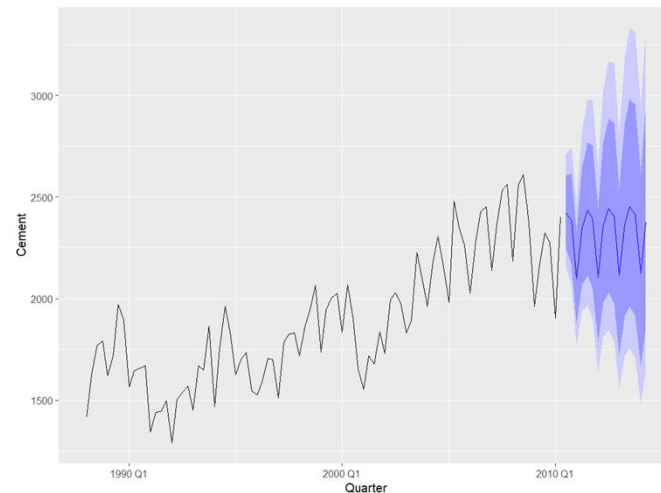
T=1000



Still a white noise. These spikes outside the bounds are not statistically significant, since  $(5\%)(30) = 1.5$

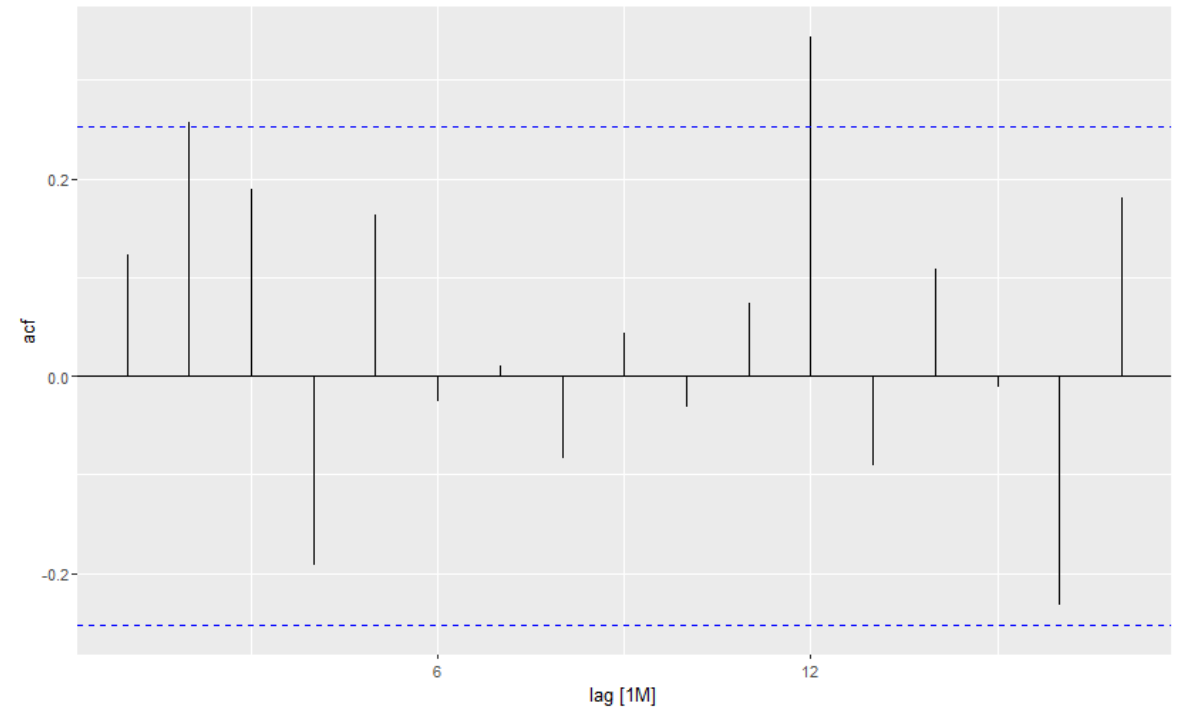
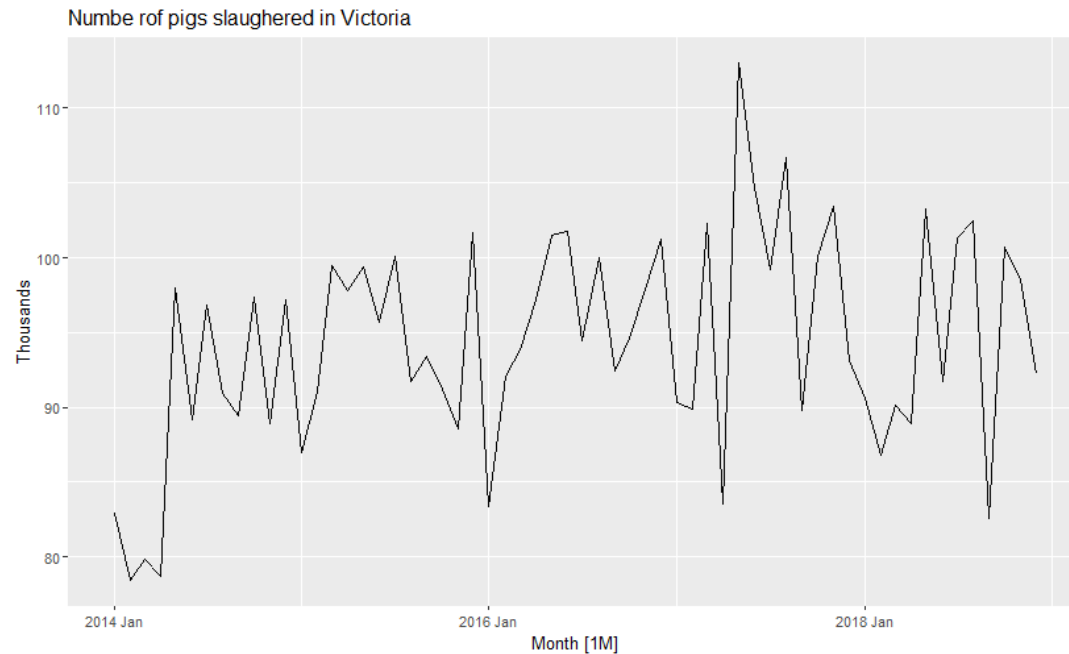
# Sampling distribution of autocorrelations

- Sampling distribution of  $r_k$  for white noise data is asymptotically  $N(0, \frac{1}{T})$  where  $T$  is the length of the time series.
- 95% of all  $r_k$  for white noise lie within  $\pm 2/\sqrt{T}$
- If this is not the case, the series is probably not white noise.
- Common to plot lines at  $\pm 2/\sqrt{T}$  when plotting ACT. These are the **critical values** (at 5% significance level).



# In-class: White noise or not?

```
pigs <- aus_livestock %>%  
  filter(State == "Victoria", Animal == "Pigs", year(Month) >= 2014)  
pigs %>% autoplot(Count / 1e3) +  
  labs(y="Thousands", title = "Number of pigs slaughtered in Victoria")  
  
pigs %>%  
  ACF(Count) %>%  
  autoplot()
```



# Portmanteau tests for autocorrelation\*

- Let  $\rho_j = \text{corr}(\epsilon_t, \epsilon_{t-j})$  be the autocorrelation at lag  $j$
- Consider a *whole set* of  $r_k$  values, and develop a test to see whether the set is significantly different from a zero set.

$$H_0: \rho_1 = \rho_2 = \dots = \rho_\ell = 0$$

- Let  $\ell$  be the maximum lag being considered
  - We suggest using  $\ell = 10$  for non-seasonal data and  $\ell = 2m$  for seasonal data, where  $m$  is the period of seasonality.

- **Box-Pierce test**  $Q = T \sum_{k=1}^{\ell} r_k^2$

- **Ljung-Box test**  $Q^* = T(T+2) \sum_{k=1}^{\ell} (T-k)^{-1} r_k^2$

- If the autocorrelations did come from a white noise series, then both  $Q$  and  $Q^*$  would have a  $\chi^2$  distribution with  $(\ell - K)$  degrees of freedom, where  $K$  is the number of parameters in the model.
  - Naïve method has no parameters, so  $K = 0$
  - Drift method has one estimated parameter (drift coefficient), so  $K = 1$

```
aug %>% features(.innov, box_pierce, lag = 10, dof = 0)
#> # A tibble: 1 × 4
#>   Symbol .model      bp_stat bp_pvalue
#>   <chr>   <chr>      <dbl>   <dbl>
#> 1 GOOG  NAIVE(Close)    7.74    0.654

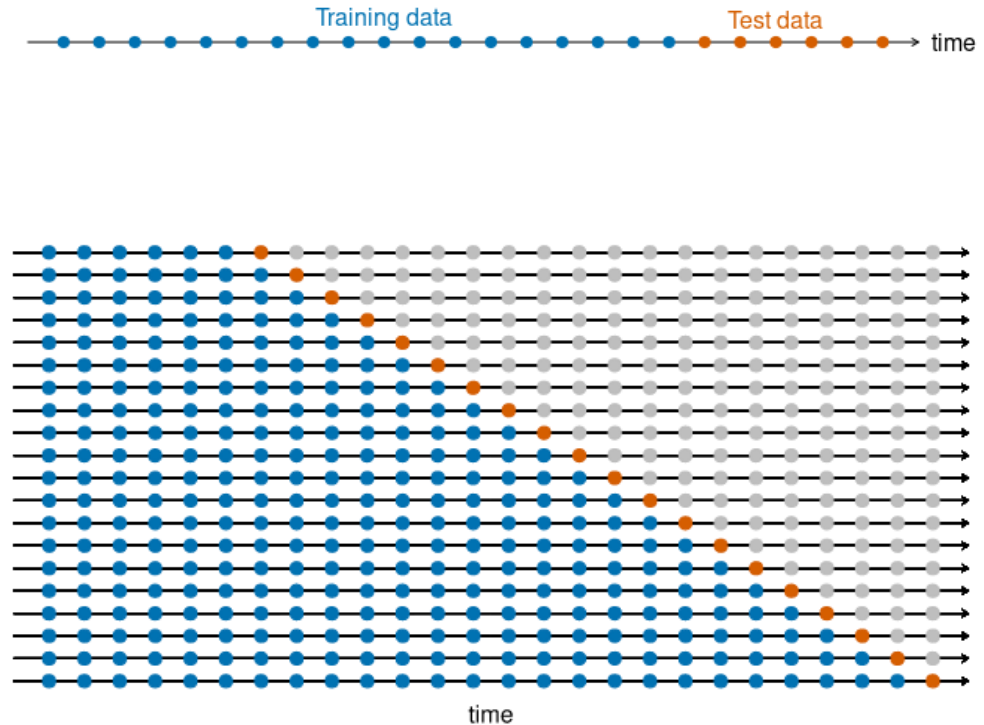
aug %>% features(.innov, ljung_box, lag = 10, dof = 0)
#> # A tibble: 1 × 4
#>   Symbol .model      lb_stat lb_pvalue
#>   <chr>   <chr>      <dbl>   <dbl>
#> 1 GOOG  NAIVE(Close)    7.91    0.637
```

Evaluate: TS cross validation

# Model evaluation methods

## Time series cross-validation

- Traditional evaluation
  - Train/test split
- **Resampling** method: Cross-validation
  - Train/test split multiple times
  - Forecast accuracy averaged over test sets
  - a.k.a. “evaluation on a rolling forecasting origin” because the “origin” at which the forecast is based rolls forward in time (“forward chaining”)



# Cross validation: Small Example StoreID =2, SKU =A

| t  | Date       | Demand<br>StoreID=2, SKU=A |
|----|------------|----------------------------|
| 1  | 2001-01-01 | 190                        |
| 2  | 2001-04-01 | 260                        |
| 3  | 2001-07-01 | 330                        |
| 4  | 2001-10-01 | 480                        |
| 5  | 2002-01-01 | 570                        |
| 6  | 2002-04-01 | 710                        |
| 7  | 2002-07-01 | 650                        |
| 8  | 2002-10-01 | 690                        |
| 9  | 2003-01-01 | 730                        |
| 10 | 2003-04-01 | 680                        |
| 11 | 2003-07-01 | 660                        |
| 12 | 2003-10-01 | 590                        |

File 02-introTS-FC  
Sheet 2A



| t          | 1       | 2       | 3       | 4       | 5       | 6       | 7       | 8       | 9       | 10      | 11      | 12      |            |         |          |  |
|------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|------------|---------|----------|--|
| Date       | 2001 Q1 | 2001 Q2 | 2001 Q3 | 2001 Q4 | 2002 Q1 | 2002 Q2 | 2002 Q3 | 2002 Q4 | 2003 Q1 | 2003 Q2 | 2003 Q3 | 2003 Q4 |            |         |          |  |
| Demand     | 190     | 260     | 330     | 480     | 570     | 710     | 650     | 690     | 730     | 680     | 660     | 590     |            |         |          |  |
| Resample # | Train   |         |         |         |         |         |         |         |         |         |         |         | Validation | FC.Mean | FC.Naive |  |
| 1          | 190     | 260     | 330     |         |         |         |         |         |         |         |         |         | 480        | 260.0   |          |  |
| 2          | 190     | 260     | 330     | 480     |         |         |         |         |         |         |         |         | 570        | 315.0   |          |  |
| 3          | 190     | 260     | 330     | 480     | 570     |         |         |         |         |         |         |         | 710        | 366.0   |          |  |
| 4          | 190     | 260     | 330     | 480     | 570     | 710     |         |         |         |         |         |         | 650        | 423.3   |          |  |
| 5          | 190     | 260     | 330     | 480     | 570     | 710     | 650     |         |         |         |         |         | 690        | 455.7   |          |  |
| 6          | 190     | 260     | 330     | 480     | 570     | 710     | 650     | 690     |         |         |         |         | 730        | 485.0   |          |  |
| 7          | 190     | 260     | 330     | 480     | 570     | 710     | 650     | 690     | 730     |         |         |         | 680        | 512.2   |          |  |
| 8          | 190     | 260     | 330     | 480     | 570     | 710     | 650     | 690     | 730     | 680     |         |         | 660        | 529.0   |          |  |
| 9          | 190     | 260     | 330     | 480     | 570     | 710     | 650     | 690     | 730     | 680     | 660     |         | 590        | 540.9   |          |  |
|            |         |         |         |         |         |         |         |         |         |         |         |         | MAE        | 208.1   |          |  |

# TS Cross validation: R vs Excel

File 02-introTS-FC

Sheet 2A

| t          | 1       | 2       | 3       | 4       | 5       | 6       | 7       | 8       | 9       | 10      | 11      | 12      |            |         |          |  |
|------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|------------|---------|----------|--|
| Date       | 2001 Q1 | 2001 Q2 | 2001 Q3 | 2001 Q4 | 2002 Q1 | 2002 Q2 | 2002 Q3 | 2002 Q4 | 2003 Q1 | 2003 Q2 | 2003 Q3 | 2003 Q4 |            |         |          |  |
| Demand     | 190     | 260     | 330     | 480     | 570     | 710     | 650     | 690     | 730     | 680     | 660     | 590     |            |         |          |  |
| Resample # | Train   |         |         |         |         |         |         |         |         |         |         |         | Validation | FC.Mean | FC.Naive |  |
| 1          | 190     | 260     | 330     |         |         |         |         |         |         |         |         |         | 480        | 260.0   |          |  |
| 2          | 190     | 260     | 330     | 480     |         |         |         |         |         |         |         |         | 570        | 315.0   |          |  |
| 3          | 190     | 260     | 330     | 480     | 570     |         |         |         |         |         |         |         | 710        | 366.0   |          |  |
| 4          | 190     | 260     | 330     | 480     | 570     | 710     |         |         |         |         |         |         | 650        | 423.3   |          |  |
| 5          | 190     | 260     | 330     | 480     | 570     | 710     | 650     |         |         |         |         |         | 690        | 455.7   |          |  |
| 6          | 190     | 260     | 330     | 480     | 570     | 710     | 650     | 690     |         |         |         |         | 730        | 485.0   |          |  |
| 7          | 190     | 260     | 330     | 480     | 570     | 710     | 650     | 690     | 730     |         |         |         | 680        | 512.2   |          |  |
| 8          | 190     | 260     | 330     | 480     | 570     | 710     | 650     | 690     | 730     | 680     |         |         | 660        | 529.0   |          |  |
| 9          | 190     | 260     | 330     | 480     | 570     | 710     | 650     | 690     | 730     | 680     | 660     |         | 590        | 540.9   |          |  |
|            |         |         |         |         |         |         |         |         |         |         |         |         | MAE        | 208.1   | 73.3     |  |

```

myts_trCV <- myts %>%
  filter(StoreID == 2, SKU == "A") %>%
  slice(1:(n()-1)) %>%
  stretch_tsibble(.init=3, .step=1) %>%
  relocate(.id, Quarter)

```

R File 02-introTS-FC.R

```

myts_trCV %>%
  model(
    Mean = MEAN(Demand),
    Naive = NAIVE(Demand)
  ) %>%
  forecast(h=1) %>%
  accuracy(myts)

```

```

# A tibble: 2 x 12
  .model StoreID SKU .type ME RMSE MAE MPE MAPE MASE RMSSE ACF1
  <chr> <dbl> <chr> <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 Mean 2 A Test 208. 223. 208. 32.7 32.7 0.934 0.817 0.442
2 Naive 2 A Test 28.9 84.9 73.3 5.17 12.2 0.329 0.311 0.197

```



# Time series cross-validation

## Evaluation on a rolling forecasting origin

- Stretch/expand: extends a growing length window with new data

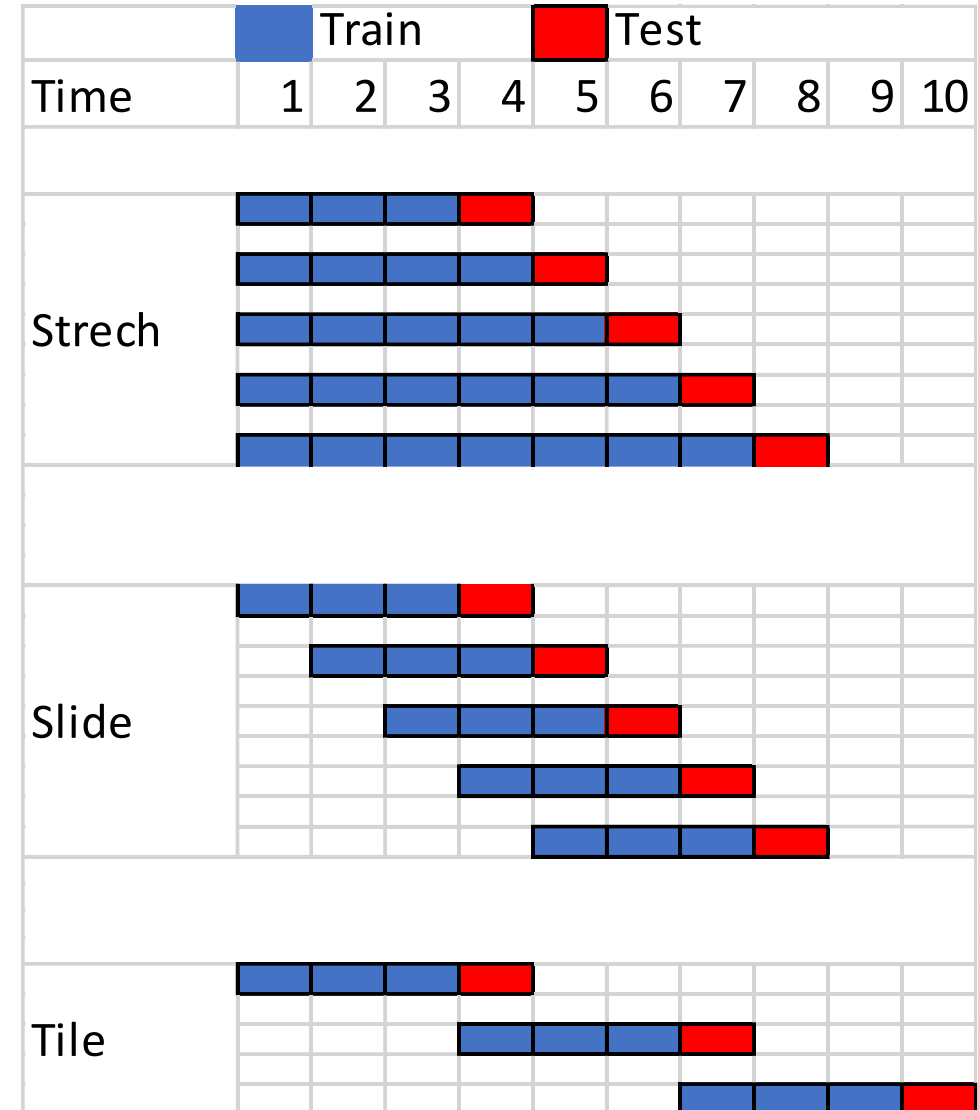
```
google_2015 %>%
  stretch_tsibble(.init = 3, .step = 1)
```

- Slide: shifts a fixed length window through data

```
google_2015 %>%
  slide_tsibble(.size = 3, .step = 1)
```

- Tile: moves a fixed length window without overlap

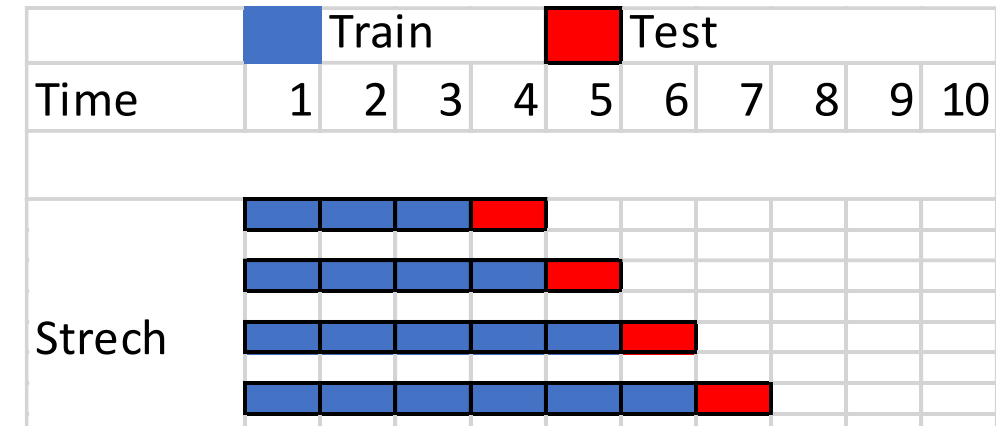
```
google_2015 %>%
  tile_tsibble(.size = 3) %>%
```



# TS Cross Validation

Small Example:  
StoreID =2, SKU =A

```
myts_trCV <- myts %>%
  filter(StoreID == 2, SKU == "A") %>%
  slice(1:(n()-1)) %>%
  stretch_tsibble(.init=3, .step=1) %>%
  relocate(.id, Quarter)
```



| t          | 1     | 2   | 3   | 4   | 5   | 6   | 7   | 8   | 9   | 10  | 11  | 12  |      |         |
|------------|-------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|---------|
| Demand     | 190   | 260 | 330 | 480 | 570 | 710 | 650 | 690 | 730 | 680 | 660 | 590 |      |         |
| Resample # | Train |     |     |     |     |     |     |     |     |     |     |     | Test | FC.Mean |
| 1          | 190   | 260 | 330 |     |     |     |     |     |     |     |     |     | 480  | 260.0   |
| 2          | 190   | 260 | 330 | 480 |     |     |     |     |     |     |     |     | 570  | 315.0   |
| 3          | 190   | 260 | 330 | 480 | 570 |     |     |     |     |     |     |     | 710  | 366.0   |
| ⋮          |       |     |     |     |     |     |     |     |     |     |     |     |      |         |
| 9          | 190   | 260 | 330 | 480 | 570 | 710 | 650 | 690 | 730 | 680 | 660 |     | 590  | 540.9   |

# of resamples is  $(12-1) - (3-1) = 9$

```
# A tsibble: 63 x 5 [1Q]
# Key:           .id, StoreID, SKU [9]
#   .id Quarter StoreID SKU   Demand
#   <int>   <qtr>   <dbl> <chr> <dbl>
1     1 2001 Q1       2 A     190
2     1 2001 Q2       2 A     260
3     1 2001 Q3       2 A     330
4     2 2001 Q1       2 A     190
5     2 2001 Q2       2 A     260
6     2 2001 Q3       2 A     330
7     2 2001 Q4       2 A     480
8     3 2001 Q1       2 A     190
9     3 2001 Q2       2 A     260
10    3 2001 Q3       2 A     330
11    3 2001 Q4       2 A     480
12    3 2002 Q1       2 A     570
13    4 2001 Q1       2 A     190
```

# ... with 50 more rows  
# i Use `print(n = ...)` to see more rows

# TS Cross validation

| t  | Date       | Demand<br>StoreID=2, SKU=A |
|----|------------|----------------------------|
| 1  | 2001-01-01 | 190                        |
| 2  | 2001-04-01 | 260                        |
| 3  | 2001-07-01 | 330                        |
| 4  | 2001-10-01 | 480                        |
| 5  | 2002-01-01 | 570                        |
| 6  | 2002-04-01 | 710                        |
| 7  | 2002-07-01 | 650                        |
| 8  | 2002-10-01 | 690                        |
| 9  | 2003-01-01 | 730                        |
| 10 | 2003-04-01 | 680                        |
| 11 | 2003-07-01 | 660                        |
| 12 | 2003-10-01 | 590                        |

| t      | 1       | 2       | 3       | 4       | 5       | 6       | 7       | 8       | 9       | 10      | 11      | 12      |          |         |          |
|--------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|----------|---------|----------|
| Date   | 2001 Q1 | 2001 Q2 | 2001 Q3 | 2001 Q4 | 2002 Q1 | 2002 Q2 | 2002 Q3 | 2002 Q4 | 2003 Q1 | 2003 Q2 | 2003 Q3 | 2003 Q4 |          |         |          |
| Demand | 190     | 260     | 330     | 480     | 570     | 710     | 650     | 690     | 730     | 680     | 660     | 590     |          |         |          |
| ID     | Train   |         |         |         |         |         |         |         |         |         |         |         | Test     | FC.Mean | FC.Naive |
| 1      | 190     | 260     | 330     |         |         |         |         |         |         |         |         |         | 480      | 260.0   |          |
| 2      | 190     | 260     | 330     | 480     |         |         |         |         |         |         |         |         | 570      | 315.0   |          |
| 3      | 190     | 260     | 330     | 480     | 570     |         |         |         |         |         |         |         | 710      | 366.0   |          |
| 4      | 190     | 260     | 330     | 480     | 570     | 710     |         |         |         |         |         |         | 650      | 423.3   |          |
| 5      | 190     | 260     | 330     | 480     | 570     | 710     | 650     |         |         |         |         |         | 690      | 455.7   |          |
| 6      | 190     | 260     | 330     | 480     | 570     | 710     | 650     | 690     |         |         |         |         | 730      | 485.0   |          |
| 7      | 190     | 260     | 330     | 480     | 570     | 710     | 650     | 690     | 730     |         |         |         | 680      | 512.2   |          |
| 8      | 190     | 260     | 330     | 480     | 570     | 710     | 650     | 690     | 730     | 680     |         |         | 660      | 529.0   |          |
| 9      | 190     | 260     | 330     | 480     | 570     | 710     | 650     | 690     | 730     | 680     | 660     |         | 590      | 540.9   |          |
|        |         |         |         |         |         |         |         |         |         |         |         |         | Test MAE | 208.1   | 73       |

```
myts_trCV <- myts %>%
  filter(StoreID == 2, SKU == "A") %>%
  slice(1:(n()-1)) %>%
  stretch_tsibble(.init=3, .step=1) %>%
  relocate(.id, Quarter)
```

```
myts_trCV %>%
  model(
    Mean = MEAN(Demand)
  ) %>%
  forecast(h=1) %>%
  accuracy(myts)
```

Try without -1

# R moment

```
df <- read_csv("smallExample.csv") # from Lecture02
head(df)
myts <- df %>%
  mutate(Quarter = yearquarter(Date) ) %>%
  as_tsibble(key = c(StoreID, SKU ), index = Quarter ) %>%
  select(-Date)
```

```
myts_trCV <- myts %>%
  filter(StoreID == 2, SKU == "A") %>%
  slice(1:(n()-1)) %>%
  stretch_tsibble(.init=3, .step=1) %>%
  relocate(.id, Quarter)
```

Note the difference between `=` and `==`

```
> tail(myts_trCV)
# A tsibble: 6 x 5 [1Q]
# Key:       .id, StoreID, SKU [1]
  .id Quarter StoreID SKU Demand
<int> <qtr> <dbl> <chr> <dbl>
1     9 2002 Q2       2  A    710
2     9 2002 Q3       2  A    650
3     9 2002 Q4       2  A    690
4     9 2003 Q1       2  A    730
5     9 2003 Q2       2  A    680
6     9 2003 Q3       2  A    660
```

```
> print(myts_trCV, n=13)
# A tsibble: 63 x 5 [1Q]
# Key:       .id, StoreID, SKU [9]
  .id Quarter StoreID SKU Demand
<int> <qtr> <dbl> <chr> <dbl>
1     1 2001 Q1       2  A    190
2     1 2001 Q2       2  A    260
3     1 2001 Q3       2  A    330
4     2 2001 Q1       2  A    190
5     2 2001 Q2       2  A    260
6     2 2001 Q3       2  A    330
7     2 2001 Q4       2  A    480
8     3 2001 Q1       2  A    190
9     3 2001 Q2       2  A    260
10    3 2001 Q3       2  A    330
11    3 2001 Q4       2  A    480
12    3 2002 Q1       2  A    570
13    4 2001 Q1       2  A    190
# ... with 50 more rows
# i Use `print(n = ...)` to see more rows
```

Evaluate: residual diagnostics

# Random error term $\epsilon_t$

- Assume
  1. [*zero bias*] Expected value of each error term is zero:  $E[\epsilon_t] = 0$
  2. [**white noise**] Errors for different time periods are independent of (or at least uncorrelated with) one another and also independent of past states
  3. [*homoskedasticity*] Variance of the errors is constant:  $var(\epsilon_t) = \sigma^2$
  4. Errors are drawn from a normal distribution
- A short-hand notation  $\epsilon_t \sim \text{NID}(0, \sigma^2)$
- NID stands for “normally and independently distributed”.

# Random *error* term $\epsilon_t$

- Assume
  1. [*zero bias*] Expected value of each error term is zero:  $E[\epsilon_t] = 0$
  2. [*white noise*] Errors for different time periods are independent of (or at least uncorrelated with) one another and also independent of past states
  3. [*homoskedasticity*] Variance of the errors is constant:  $var(\epsilon_t) = \sigma^2$
  4. Errors are drawn from a normal distribution
- A short-hand notation  $\epsilon_t \sim \text{NID}(0, \sigma^2)$
- NID stands for “normally and independently distributed”.
- **Residual analysis** checks these assumptions.

# Terminology\*

## Residuals vs Errors

Training

Test

- Training data  $\{y_1, y_2, \dots, y_T\}$
- Test data  $\{y_{T+1}, y_{T+2}, \dots\}$
- A forecast error is the difference between an observed value and its forecast:

$$e_{T+h} = y_{T+h} - \hat{y}_{T+h|T}$$

- Forecast errors are different from residuals in two ways.

- **residuals** = actual – **fitted** on the *training* set, while  
**errors** = actual – **forecast** on the *test* set.

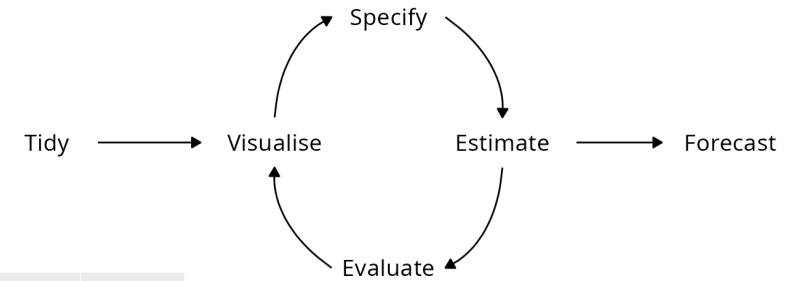
- Residuals are based on *one-step* forecasts, while errors can involve *multi-step* forecasts.

| Set   | t  | Date       | Actual | Naïve | Residual | Error |
|-------|----|------------|--------|-------|----------|-------|
| Train | 1  | 2001-01-01 | 190    |       |          |       |
| Train | 2  | 2001-04-01 | 260    | 190   | 70       |       |
| Train | 3  | 2001-07-01 | 330    | 260   | 70       |       |
| Train | 4  | 2001-10-01 | 480    | 330   | 150      |       |
| Train | 5  | 2002-01-01 | 570    | 480   | 90       |       |
| Train | 6  | 2002-04-01 | 710    | 570   | 140      |       |
| Train | 7  | 2002-07-01 | 650    | 710   | -60      |       |
| Train | 8  | 2002-10-01 | 690    | 650   | 40       |       |
| Test  | 9  | 2003-01-01 | 730    | 690   |          | 40    |
| Test  | 10 | 2003-04-01 | 680    | 690   |          | -10   |
| Test  | 11 | 2003-07-01 | 660    | 690   |          | -30   |
| Test  | 12 | 2003-10-01 | 590    | 690   |          | -100  |



# Forecasting workflow:

## Residual diagnostics



```

#####
# Example 1: global economy
# tidy
gdppc <- global_economy |>
  mutate(GDP_per_capita = GDP / Population)

# visualize
gdppc |>
  filter(Country == "Sweden") |>
  autoplot(GDP_per_capita) +
  labs(y = "$US", title = "GDP per capita for Sweden")

# specify
TSLM(GDP_per_capita ~ trend())

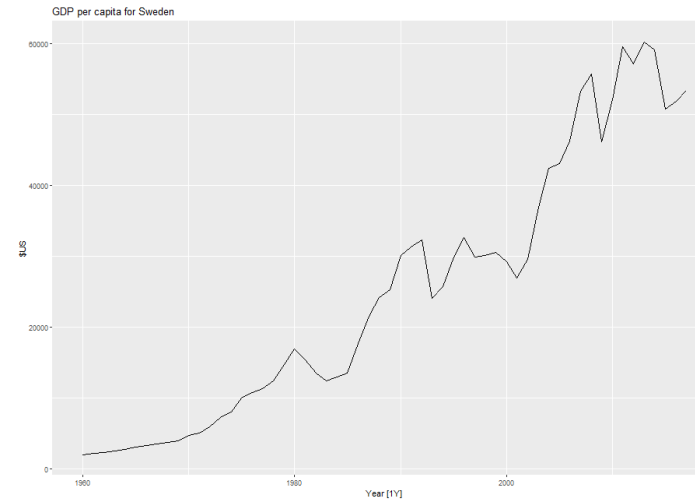
# estimate
fit <- gdppc |>
  model(trend_model = TSLM(GDP_per_capita ~ trend()))

fit

fit %>%
  filter(Country == "Sweden") %>%
  report()

# evaluate
fit %>%
  filter(Country == "Sweden") %>%
  gg_tsresiduals()

# forecast
fit |>
  forecast(h = "3 years") |>
  filter(Country == "Sweden") |>
  autoplot(gdppc) +
  labs(y = "$US", title = "GDP per capita for Sweden")
  
```



Series: GDP\_per\_capita  
Model: TSLM

Residuals:

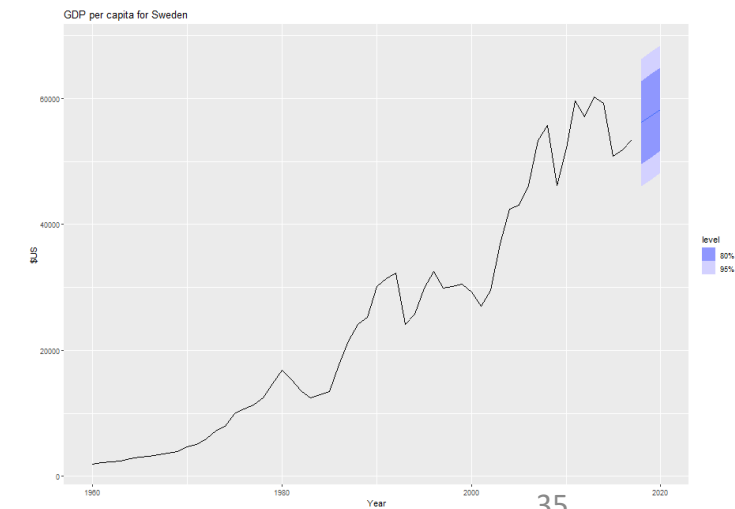
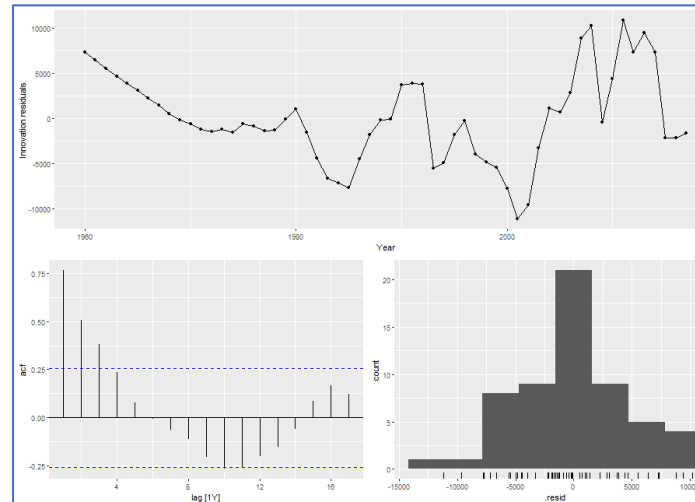
|  | Min      | 1Q      | Median | 3Q     | Max     |
|--|----------|---------|--------|--------|---------|
|  | -11170.4 | -2193.1 | -505.7 | 3524.9 | 10850.2 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | -6394.78 | 1324.72    | -4.827  | 1.11e-05 *** |
| trend()     | 1060.34  | 39.06      | 27.150  | < 2e-16 ***  |

---  
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4979 on 56 degrees of freedom  
Multiple R-squared: 0.9294, Adjusted R-squared: 0.9281  
F-statistic: 737.1 on 1 and 56 DF, p-value: < 2.22e-16



# Homework

# Hw: Cross validation

- Dataset `aus_production`. Use data from 1988 Q1 onward.
- Consider only the forecasting methods shown in the code snippet.
- Cross-validation setup: Perform TS cross validation with:
  - Step size: 1 quarter
  - Forecast horizon: 1 quarter ahead
  - Initial training window: 2 years
- The resampling scheme is expanding-window.
  - Resample#1: training 1988Q1-1989Q4; validation 1990Q1
  - Resample#2: training 1988Q1-1990Q1; validation 1990Q2
  - ...
  - Continue until no further 1-step-ahead validation observation is available.
  - Total # of resamples is \_\_\_\_\_
- Model evaluation
  - For each method: Report the average RMSE across all samples.
  - Select the best forecasting method based on lowest cross-validated RMSE.

```
my_cement <- aus_production %>%  
  filter(year(Quarter) >= 1988) %>%  
  select(Cement)  
  
myts <- my_cement  
  
myts_trCV <- myts %>%  
  slice(1:(n()-1)) %>%  
  stretch_tsibble(.init = xx, .step=yy)  
  
myts_trCV %>%  
  model(  
    Mean = MEAN(Cement),  
    Naive = NAIVE(Cement),  
    SNaive = SNAIVE(Cement),  
    Drift = NAIVE(Cement~drift()),  
    Ets = ETS(Cement),  
    Arima = ARIMA(Cement)  
  ) %>%  
  forecast(h= zz) %>%  
  accuracy(myts)
```

# Appendix

# Backshift notations

## Differencing

# Stationarity & differencing

- A **stationary** time series is one whose properties do not depend on the time at which the series is observed
- Log transformations can help to stabilize the variance of a time series. Differencing helps to stabilize the mean.
- The **differenced series** is the *change* between each observation in the original series:

$$y'_t = y_t - y_{t-1}$$

- Occasionally the differenced data will not appear stationary and it may be necessary to difference the data a second time.

**second-order differencing:**  $y''_t = y'_t - y'_{t-1}$

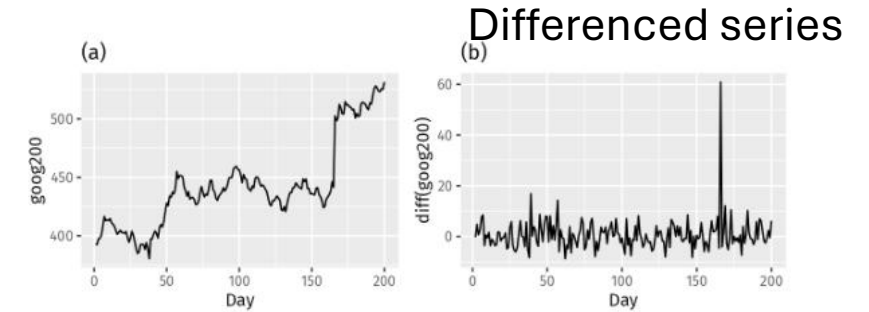
- In practice, it is almost never necessary to go beyond second-order differences

- **Seasonal differencing**, a.k.a., **lag-m differences**

$$y'_t = y_t - y_{t-m}$$

where  $m$  is the seasonal period (e.g.,  $m = 4$  for quarterly data,  $m = 12$  for monthly data)

See details in Appendix



```
cbind("Sales ($million)" = a10,  
      "Monthly log sales" = log(a10),  
      "Annual change in log sales" = diff(log(a10),12)) %>%  
autoplot(facets=TRUE) +  
  xlab("Year") + ylab("") +  
  ggtitle("Antidiabetic drug sales")
```

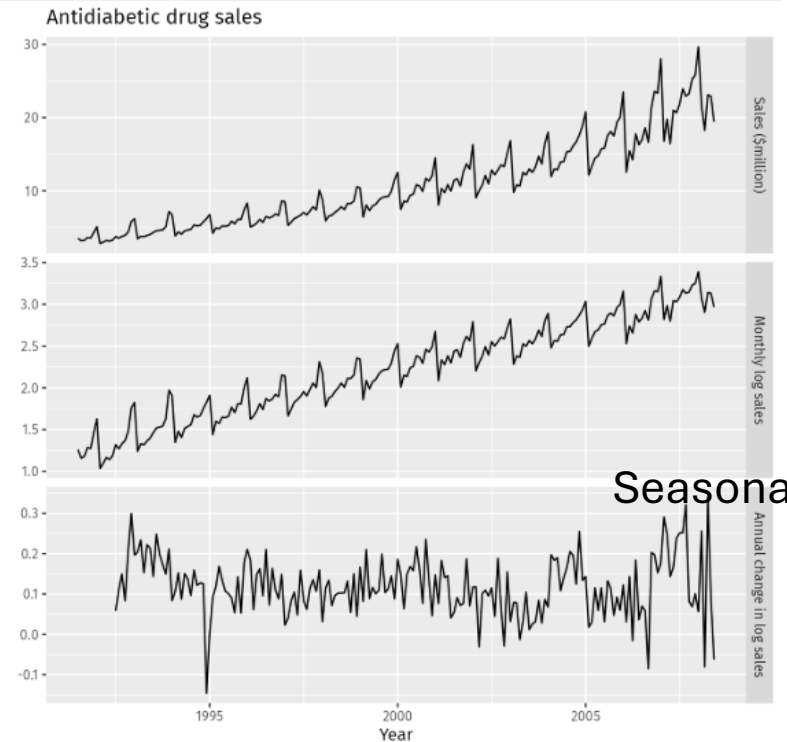


Figure 8.3: Logs and seasonal differences of the A10 (antidiabetic) sales data. The logarithms stabilise the variance, while the seasonal differences remove the seasonality and trend.

# Backshift notation $B^m y_t = y_{t-m}$

- $By_t = y_{t-1}$  Shifting data back one period
- $B^2 y_t = B(By_t) = By_{t-1} = y_{t-2}$
- For monthly data, if we wish to consider “the same month last year”, the notation is \_\_\_\_\_
- Some references use  $L$  for “lag” instead of  $B$  for backshift.

| Quarter | $t$ | Yr | Qtr | Observation<br>$y_t$ | Lag1<br>$By_t = y_{t-1}$ | Lag2<br>$B^2 y_t = y_{t-2}$ |
|---------|-----|----|-----|----------------------|--------------------------|-----------------------------|
| 2000 Q1 | 1   | 1  | 1   | 421                  |                          |                             |
| 2000 Q2 | 2   | 1  | 2   | 402                  | 421                      |                             |
| 2000 Q3 | 3   | 1  | 3   | 414                  | 402                      | 421                         |
| 2000 Q4 | 4   | 1  | 4   | 500                  | 414                      | 402                         |
| 2001 Q1 | 5   | 2  | 1   | 451                  | 500                      | 414                         |
| 2001 Q2 | 6   | 2  | 2   | 380                  | 451                      | 500                         |
| 2001 Q3 | 7   | 2  | 3   | 416                  | 380                      | 451                         |
| 2001 Q4 | 8   | 2  | 4   | 492                  | 416                      | 380                         |
| 2002 Q1 | 9   | 3  | 1   | 428                  | 492                      | 416                         |
| 2002 Q2 | 10  | 3  | 2   | 408                  | 428                      | 492                         |
| 2002 Q3 | 11  | 3  | 3   | 406                  | 408                      | 428                         |
| 2002 Q4 | 12  | 3  | 4   | 506                  | 406                      | 408                         |
| 2003 Q1 | 13  | 4  | 1   | 435                  | 506                      | 406                         |
| 2003 Q2 | 14  | 4  | 2   | 380                  | 435                      | 506                         |
| 2003 Q3 | 15  | 4  | 3   | 421                  | 380                      | 435                         |
| 2003 Q4 | 16  | 4  | 4   | 490                  | 421                      | 380                         |

For quarterly data,

- Lag 4 means \_\_\_\_\_
- Lag 8 means \_\_\_\_\_

# Differencing

- $y_t - y_{t-1} = y_t - By_t = (1 - B)y_t$
- $y_t - y_{t-2} = (1 - B^2)y_t$

**first difference**

**second difference**

| Quarter | $t$ | Yr | Qtr | Observation<br>$y_t$ | Lag1<br>$y_{t-1}$ | Lag2 | First difference<br>$y_t - y_{t-1} = (1 - B)y_t$ | 2 <sup>nd</sup> difference<br>$y_t - y_{t-2} = (1 - B^2)y_t$ | 4 <sup>th</sup> difference<br>(#08) |
|---------|-----|----|-----|----------------------|-------------------|------|--------------------------------------------------|--------------------------------------------------------------|-------------------------------------|
| 2000 Q1 | 1   | 1  | 1   | 421                  |                   |      |                                                  |                                                              |                                     |
| 2000 Q2 | 2   | 1  | 2   | 402                  | 421               |      | -19                                              |                                                              |                                     |
| 2000 Q3 | 3   | 1  | 3   | 414                  | 402               | 421  | 12                                               |                                                              |                                     |
| 2000 Q4 | 4   | 1  | 4   | 500                  | 414               | 402  | 86                                               |                                                              |                                     |
| 2001 Q1 | 5   | 2  | 1   | 451                  | 500               | 414  | -49                                              |                                                              |                                     |
| 2001 Q2 | 6   | 2  | 2   | 380                  | 451               | 500  | -71                                              |                                                              |                                     |
| 2001 Q3 | 7   | 2  | 3   | 416                  | 380               | 451  | 36                                               |                                                              |                                     |
| 2001 Q4 | 8   | 2  | 4   | 492                  | 416               | 380  | 76                                               |                                                              |                                     |
| 2002 Q1 | 9   | 3  | 1   | 428                  | 492               | 416  | -64                                              |                                                              |                                     |



# $d$ th-order difference

- 2<sup>nd</sup> difference vs 2<sup>nd</sup>-order difference

- 1<sup>st</sup> difference

$$y_t - y_{t-1} = y_t - By_t = (1 - B)y_t := y'_t$$

- 2<sup>nd</sup> difference

$$y_t - y_{t-2} = (1 - B^2)y_t$$

- 2<sup>nd</sup>-order difference

$$y''_t = y'_t - y'_{t-1}$$

$$= y_t - 2y_{t-1} + y_{t-2}$$

$$= (1 - 2B + B^2)y_t$$

$$= (1 - B)^2 y_t$$

$$:= y_t^{(2)}$$

- $d$ th-order difference

$$y_t^{(d)} = (1 - B)^d y_t$$

$$y''_t = (1 - B)^2 y_t = (1 - B)y'_t = (1 - B)(1 - B)y_t$$

| Quarter | $t$ | Yr | Qtr | Observation<br>$y_t$ | Lag1<br>$By_t$ | Lag2<br>$B^2 y_t$ | First difference<br>$y'_t = (1 - B)y_t$ | 2nd difference<br>$y''_t = y'_t - y'_{t-1}$ | $y'_{t-1}$ | 2nd-order difference<br>$y''_t = y'_t - y'_{t-1}$<br>$= y_t - 2y_{t-1} + y_{t-2}$ |
|---------|-----|----|-----|----------------------|----------------|-------------------|-----------------------------------------|---------------------------------------------|------------|-----------------------------------------------------------------------------------|
| 2000 Q1 | 1   | 1  | 1   | 421                  |                |                   |                                         |                                             |            |                                                                                   |
| 2000 Q2 | 2   | 1  | 2   | 402                  | 421            |                   | -19                                     |                                             |            |                                                                                   |
| 2000 Q3 | 3   | 1  | 3   | 414                  | 402            | 421               | 12                                      | -7                                          | -19        | 31                                                                                |
| 2000 Q4 | 4   | 1  | 4   | 500                  | 414            | 402               | 86                                      | 98                                          | 12         | 74                                                                                |
| 2001 Q1 | 5   | 2  | 1   | 451                  | 500            | 414               | -49                                     | 37                                          | 86         | -135                                                                              |
| 2001 Q2 | 6   | 2  | 2   | 380                  | 451            | 500               | -71                                     | -120                                        | -49        | -22                                                                               |
| 2001 Q3 | 7   | 2  | 3   | 416                  | 380            | 451               | 36                                      | -35                                         | -71        | 107                                                                               |
| 2001 Q4 | 8   | 2  | 4   | 492                  | 416            | 380               | 76                                      | 112                                         | 36         | 40                                                                                |
| 2002 Q1 | 9   | 3  | 1   | 428                  | 492            | 416               | -64                                     | 12                                          | 76         | -140                                                                              |
| 2002 Q2 | 10  | 3  | 2   | 408                  | 428            | 492               | -20                                     | -84                                         | -64        | 44                                                                                |
| 2002 Q3 | 11  | 3  | 3   | 406                  | 408            | 428               | -2                                      | -22                                         | -20        | 18                                                                                |
| 2002 Q4 | 12  | 3  | 4   | 506                  | 406            | 408               | 100                                     | 98                                          | -2         | 102                                                                               |

# Differencing

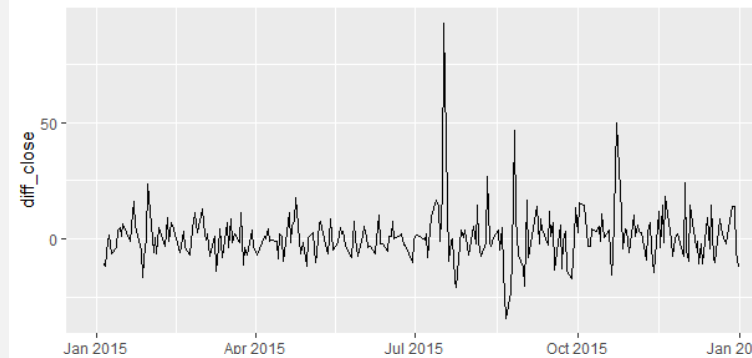
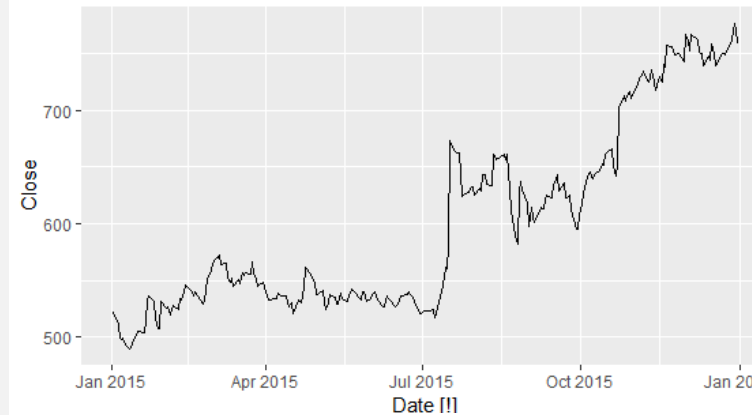
```
google_2015 <- gafa_stock %>%
  filter(Symbol == "GOOG", year(Date) == 2015)
google_2015 %>% autoplot(Close)

google_2015 %>% ACF(Close) %>%
  autoplot() + labs(subtitle = "Google closing
stock price")

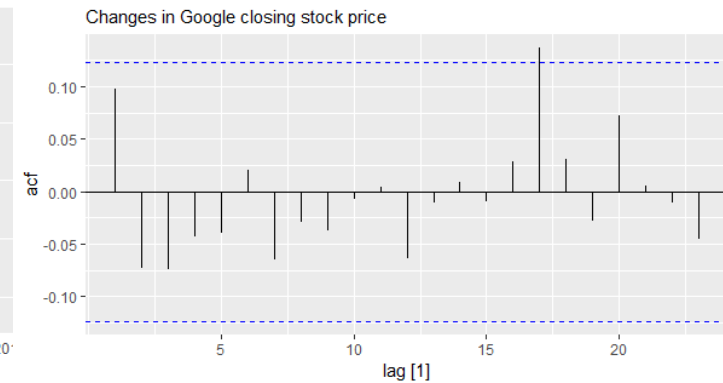
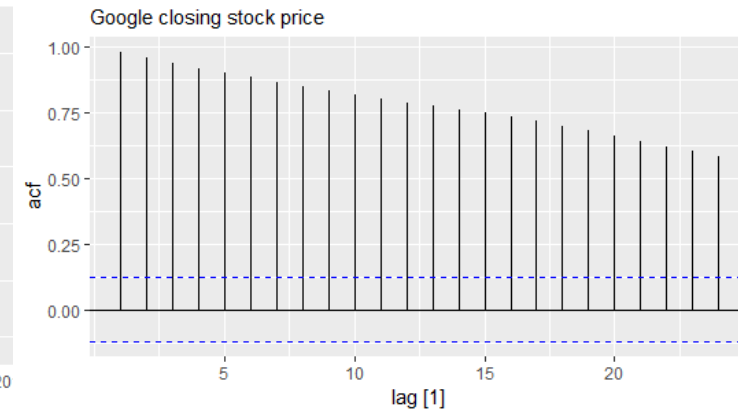
google_2015 %>% ACF(difference(Close)) %>%
  autoplot() + labs(subtitle = "Changes in Google
closing stock price")

google_2015 %>%
  mutate(diff_close = difference(Close)) %>%
  autoplot(diff_close)

google_2015 %>%
  mutate(diff_close = difference(Close)) %>%
  features(diff_close, ljung_box, lag = 10)
```



Warning message:  
Removed 1 row containing missing values (`geom\_line()`).



```
# A tibble: 1 x 3
  symbol lb_stat lb_pvalue
  <chr>   <dbl>   <dbl>
1 GOOG    7.91    0.637
```

# Seasonal difference

- $By_t = y_{t-1}$
- $B^2y_t = y_{t-2}$
- $y_t - y_{t-1} = y_t - By_t = (1 - B)y_t := y'_t$  **the first difference**
- $y_t - y_{t-2} = (1 - B^2)y_t$  the **second difference**

The **second-order difference**

$$\begin{aligned} y''_t &= y'_t - y'_{t-1} = y_t - 2y_{t-1} + y_{t-2} \\ &= (1 - 2B + B^2)y_t = (1 - B)^2y_t \end{aligned}$$

This is not the same as the second difference  $(1 - B^2)y_t$

- Let  $m$  be the seasonal period
- A **seasonal difference** is the difference between an observation and the corresponding observation from the previous year.

$$(1 - B^m)y_t$$

- Quarterly: seasonal period  $m = 4$ .  
Seasonal difference  $y_t - y_{t-4} = (1 - B^4)y_t$

- Monthly:  
**Seasonal difference**

- A **seasonal difference** followed by a first difference can be written as

$$(1 - B)(1 - B^m)y_t$$

The backshift operator  $B$  can be treated using ordinary algebraic rules.

$$\begin{aligned} (1 - B)(1 - B^m)y_t &= (1 - B - B^m + B^{m+1})y_t \\ &= y_t - y_{t-1} - y_{t-m} + y_{t-m-1} \end{aligned}$$

|         | $t$ | $y_t$ | $B^4 y_t$ | $(1 - B^4)y_t$ | $B(1 - B^4)y_t$ | $(1 - B)(1 - B^4)y_t$ |
|---------|-----|-------|-----------|----------------|-----------------|-----------------------|
| 2000 Q1 | 1   | 421   |           |                |                 |                       |
| 2000 Q2 | 2   | 402   |           |                |                 |                       |
| 2000 Q3 | 3   | 414   |           |                |                 |                       |
| 2000 Q4 | 4   | 500   |           |                |                 |                       |
| 2001 Q1 | 5   | 451   | 421       | 30             |                 |                       |
| 2001 Q2 | 6   | 380   | 402       | -22            | 30              | -52                   |
| 2001 Q3 | 7   | 416   | 414       | 2              | -22             | 24                    |
| 2001 Q4 | 8   | 492   | 500       | -8             | 2               | -10                   |
| 2002 Q1 | 9   | 428   | 451       | -23            | -8              | -15                   |
| 2002 Q2 | 10  | 408   | 380       | 28             | -23             | 51                    |
| 2002 Q3 | 11  | 406   | 416       | -10            | 28              | -38                   |
| 2002 Q4 | 12  | 506   | 492       | 14             | -10             | 24                    |

- A seasonal difference followed by a first difference can be written as

$$(1 - B)(1 - B^m)y_t = (1 - B^m)y_t - B(1 - B^m)y_t$$

- The backshift operator  $B$  can be treated using ordinary algebraic rules.

$$(1 - B)(1 - B^m)y_t = (1 - B - B^m + B^{m+1})y_t = y_t - y_{t-1} - y_{t-m} + y_{t-m-1}$$

$$\text{ARIMA}(0, \textcolor{red}{1}, 0)(0, \textcolor{blue}{1}, 0)_m: \quad (1 - B)^{\textcolor{red}{1}}(1 - B^m)^{\textcolor{blue}{1}}y_t = c + \epsilon_t$$

In the above table,  $m = 4$

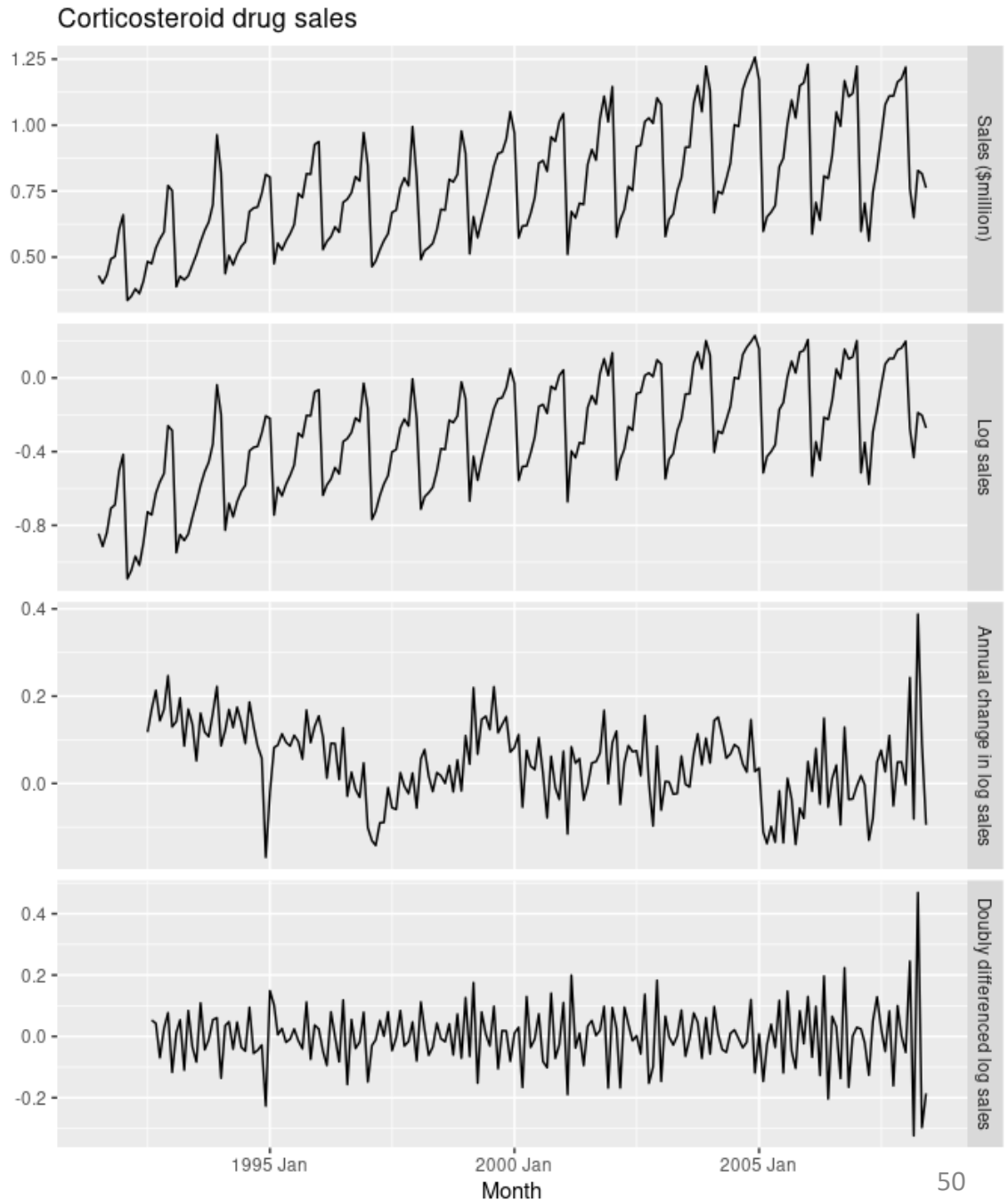
# Example

If  $y'_t = y_t - y_{t-m}$  denotes a seasonally differenced series, then the twice-differenced series is

$$\begin{aligned} y''_t &= y'_t - y'_{t-1} \\ &= (y_t - y_{t-m}) - (y_{t-1} - y_{t-m-1}) \\ &= y_t - y_{t-1} - y_{t-m} + y_{t-m-1} \end{aligned}$$

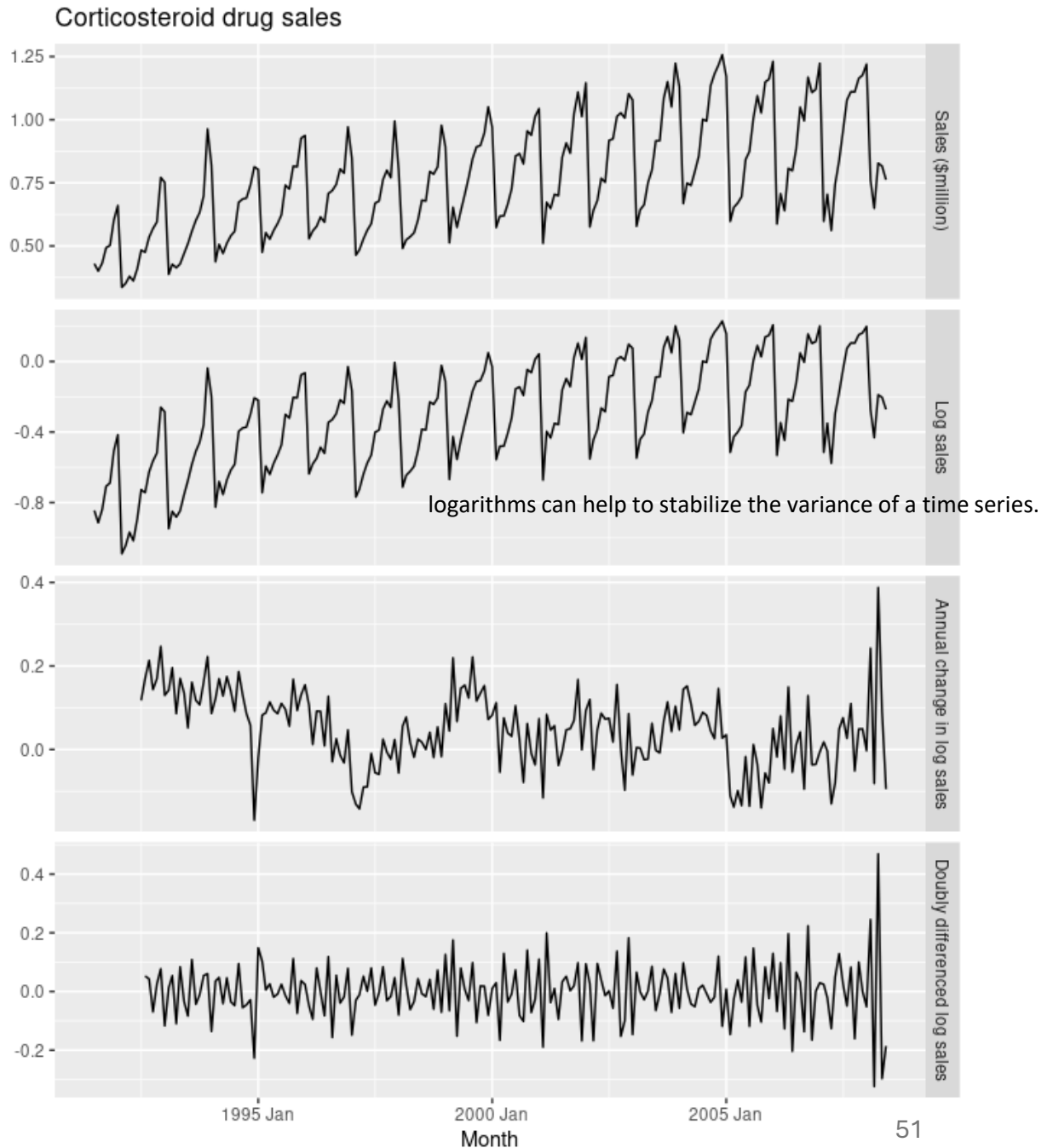
For monthly data,

$$\begin{aligned} y_t^* &= y'_t - y'_{t-1} \\ &= (y_t - y_{t-12}) - (y_{t-1} - y_{t-13}) \\ &= y_t - y_{t-1} - y_{t-12} + y_{t-13} . \end{aligned}$$



# differencing

- When both seasonal and first differences are applied, it makes no difference which is done first—the result will be the same.
- However, if the data have a strong seasonal pattern, we recommend that seasonal differencing be done first, because the resulting series will sometimes be stationary and there will be no need for a further first difference.
- If first differencing is done first, there will still be seasonality present.



ACF details

# First-order sample autocorrelation

Recall sample autocorrelation at lag  $k$

$$r_k = \frac{\sum_{t=k+1}^T (y_t - \bar{y})(y_{t-k} - \bar{y})}{\sum_{t=1}^T (y_t - \bar{y})^2}$$

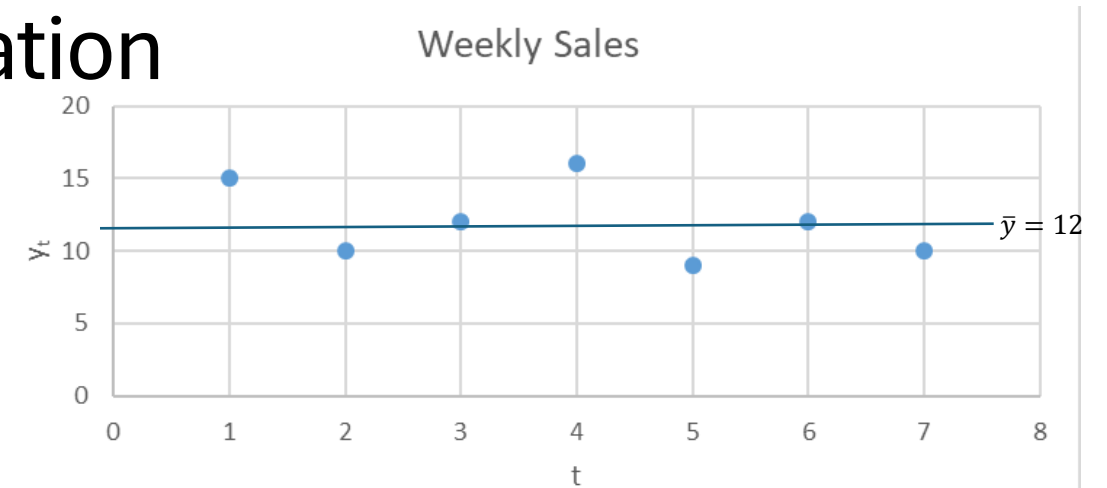
Sample autocorrelation at lag 1

$$r_1 =$$

$$r_1 = -0.429$$

Negative autocorrelation at lag 1

Interpretation: Above-average sales one week tend to be followed by somewhat below-average sales the next week.



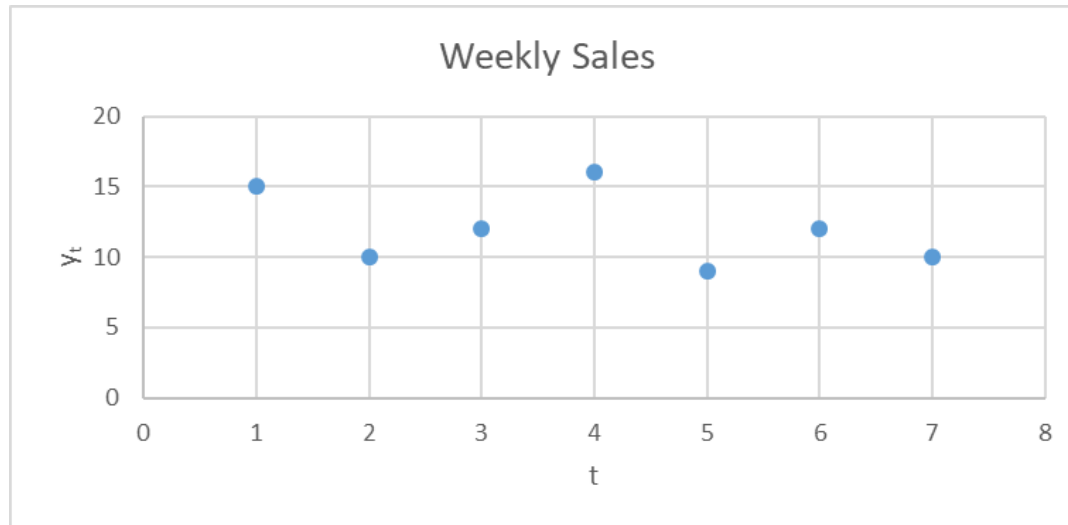
| Week<br>t      | Sales<br>y <sub>t</sub> | Lag1<br>y <sub>t-1</sub>                     | y <sub>t</sub> - $\bar{y}$ | y <sub>t-1</sub> - $\bar{y}$ | (y <sub>t</sub> - $\bar{y}$ )(y <sub>t-1</sub> - $\bar{y}$ ) | (y <sub>t</sub> - $\bar{y}$ ) <sup>2</sup> |
|----------------|-------------------------|----------------------------------------------|----------------------------|------------------------------|--------------------------------------------------------------|--------------------------------------------|
| 1              | 15                      |                                              | 3                          |                              |                                                              | 9                                          |
| 2              | 10                      | 15                                           | -2                         | 3                            | -6                                                           | 4                                          |
| 3              | 12                      | 10                                           | 0                          | -2                           | 0                                                            | 0                                          |
| 4              | 16                      | 12                                           | 4                          | 0                            | 0                                                            | 16                                         |
| 5              | 9                       | 16                                           | -3                         | 4                            | -12                                                          | 9                                          |
| 6              | 12                      | 9                                            | 0                          | -3                           | 0                                                            | 0                                          |
| 7              | 10                      | 12                                           | -2                         | 0                            | 0                                                            | 4                                          |
| SUM            |                         |                                              |                            |                              | -18                                                          | 42                                         |
| $\bar{y}$      | 12                      | $\bar{y} = \frac{15+10+12+\dots+10}{7} = 12$ |                            |                              |                                                              |                                            |
| r <sub>1</sub> | -0.429                  |                                              |                            |                              |                                                              |                                            |



# Example

| Week<br>$t$ | Sales<br>$y_t$ | Lag1<br>$y_{t-1}$ |
|-------------|----------------|-------------------|
| 1           | 15             |                   |
| 2           | 10             | 15                |
| 3           | 12             | 10                |
| 4           | 16             | 12                |
| 5           | 9              | 16                |
| 6           | 12             | 9                 |
| 7           | 10             | 12                |

Time plot



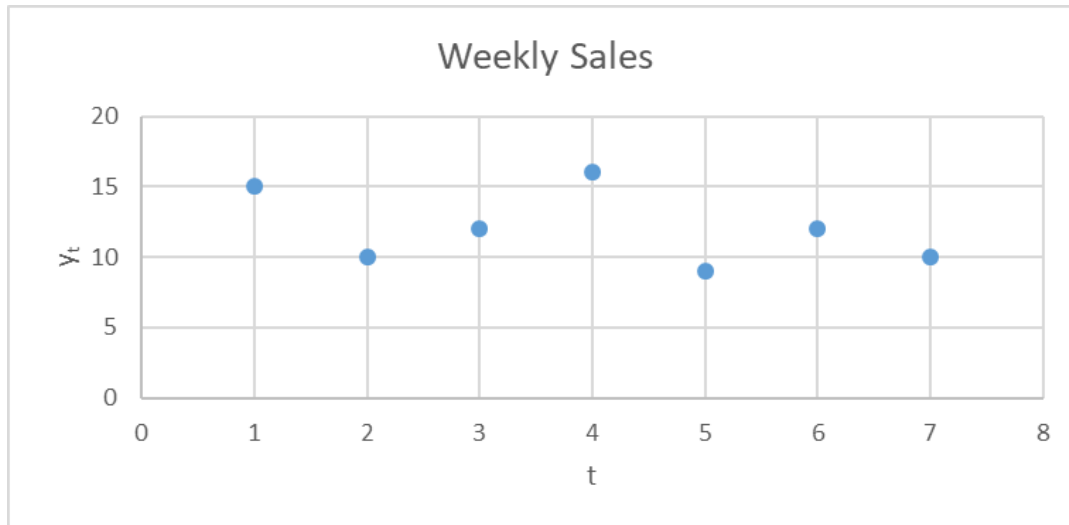
From time plot, positive or negative relationship?  
Lag plot helps to visualize the relationship

Relationship between sales in consecutive weeks: Positive or negative?

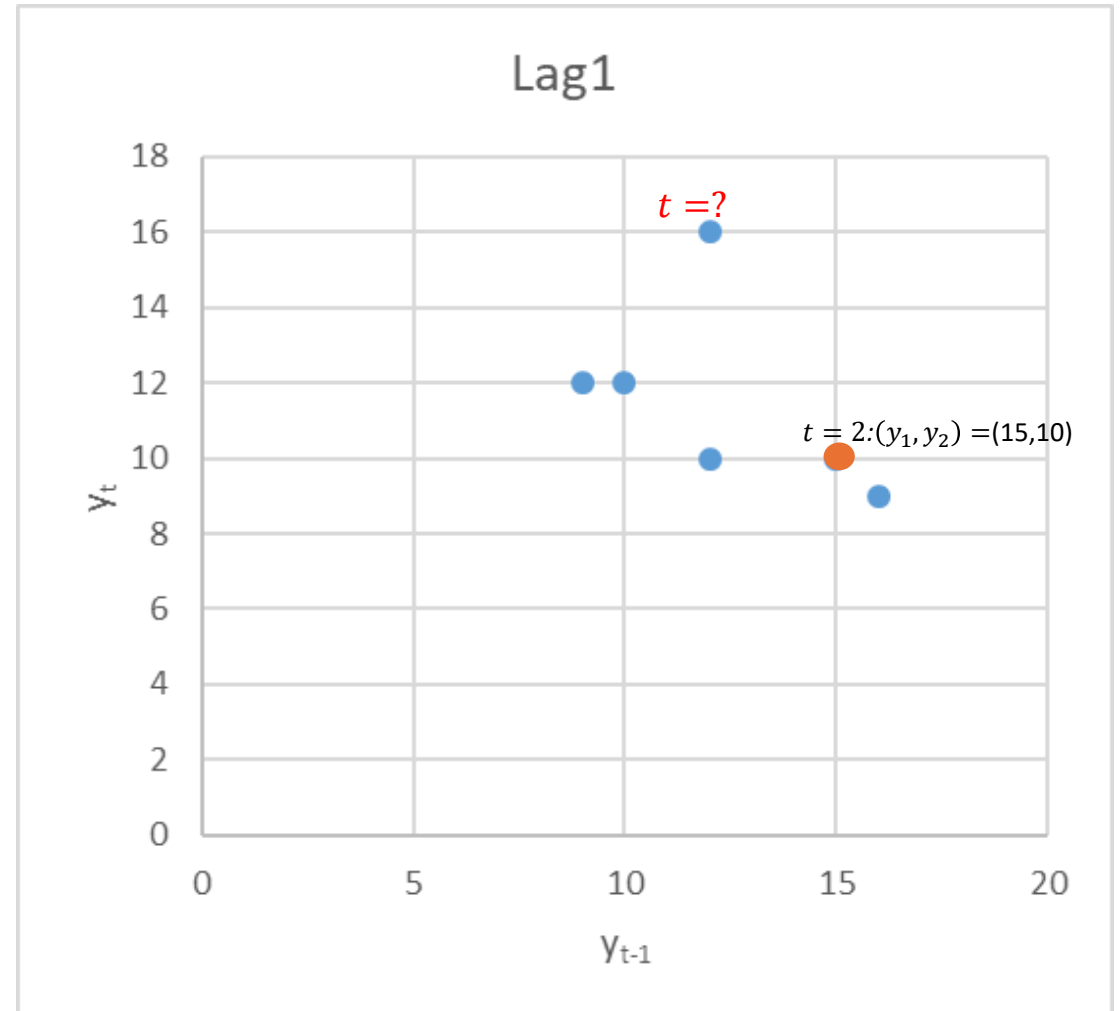
# Example

| Week<br>$t$ | Sales<br>$y_t$ | Lag1<br>$y_{t-1}$ |
|-------------|----------------|-------------------|
| 1           | 15             |                   |
| 2           | 10             | 15                |
| 3           | 12             | 10                |
| 4           | 16             | 12                |
| 5           | 9              | 16                |
| 6           | 12             | 9                 |
| 7           | 10             | 12                |

Time plot



Lag plots



# Autocorrelation function (ACF)

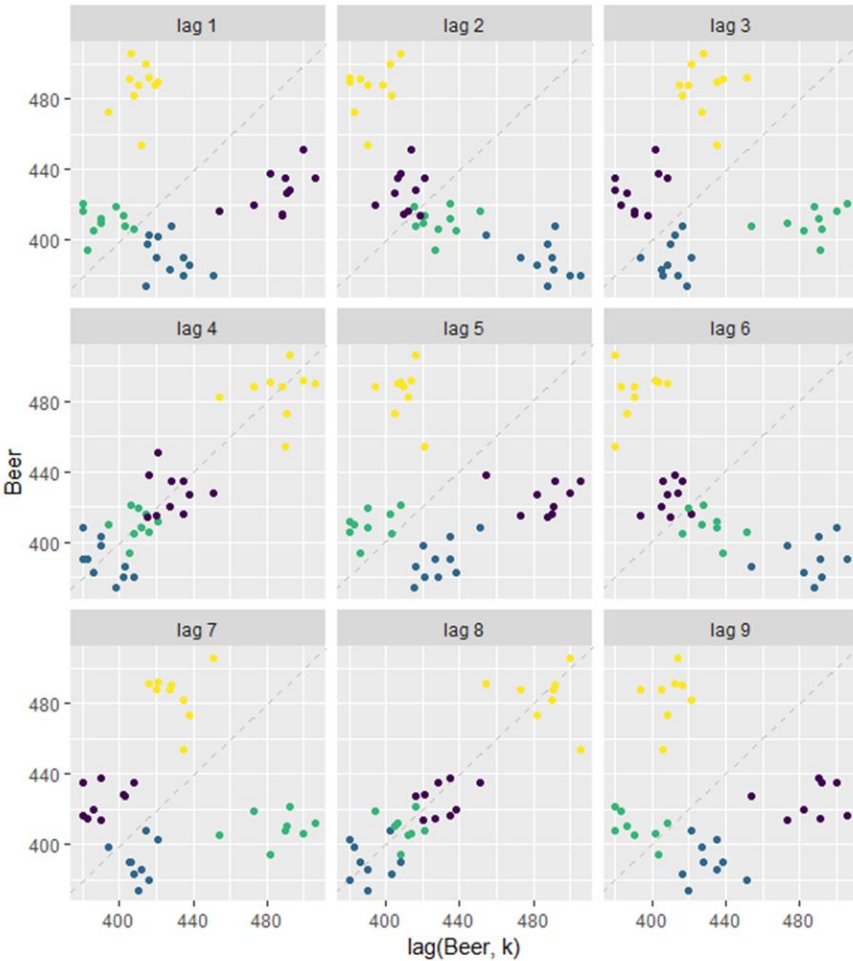
- The autocorrelation coefficients make up the **autocorrelation function (ACF)**.
- $r_1$  measures relationship between  $y_t$  and  $y_{t-1}$
- $r_2$  measures relationship between  $y_t$  and  $y_{t-2}$
- ...

```
y <- tsibble(  
  Wk = 1:7,  
  Sales = c(15, 10, 12, 16, 9, 12, 10),  
  index = Wk  
)  
  
y %>% ACF(Sales, lag_max = 4)
```

Optional: Usually no further than  $T/4$

|   | lag   | acf     |
|---|-------|---------|
|   | <lag> | <dbl>   |
| 1 | 1     | -0.429  |
| 2 | 2     | -0.0476 |
| 3 | 3     | 0.238   |
| 4 | 4     | -0.214  |

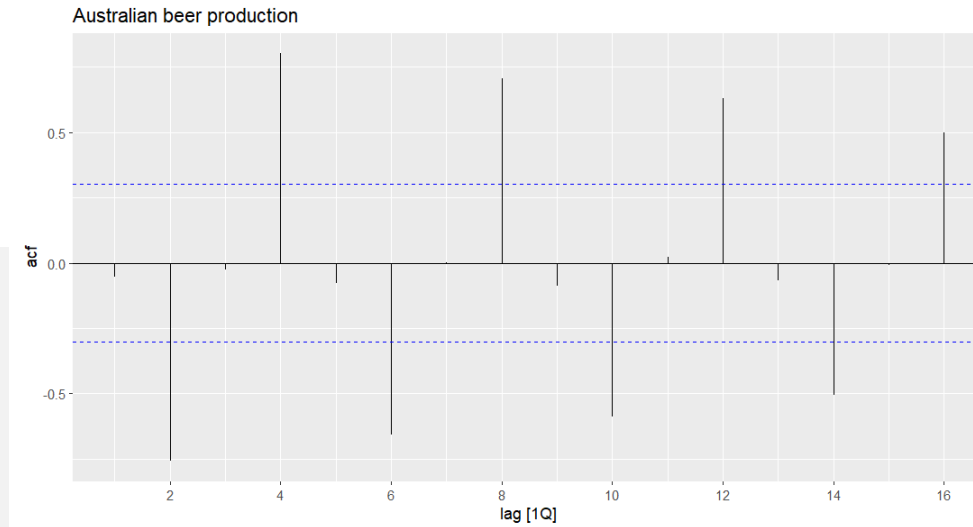
# ACF



```
autoplot(recent_production, Beer)
```

```
recent_production %>%  
  gg_lag(Beer, geom = "point") +  
  labs(x = "lag(Beer, k)")
```

```
recent_production %>% ACF(Beer, lag_max = 9)
```



|   | lag   | acf     |
|---|-------|---------|
|   | <lag> | <dbl>   |
| 1 | 1Q    | -0.0530 |
| 2 | 2Q    | -0.758  |
| 3 | 3Q    | -0.0262 |
| 4 | 4Q    | 0.802   |
| 5 | 5Q    | -0.0775 |
| 6 | 6Q    | -0.657  |
| 7 | 7Q    | 0.00119 |
| 8 | 8Q    | 0.707   |
| 9 | 9Q    | -0.0888 |

$r_1$  near zero

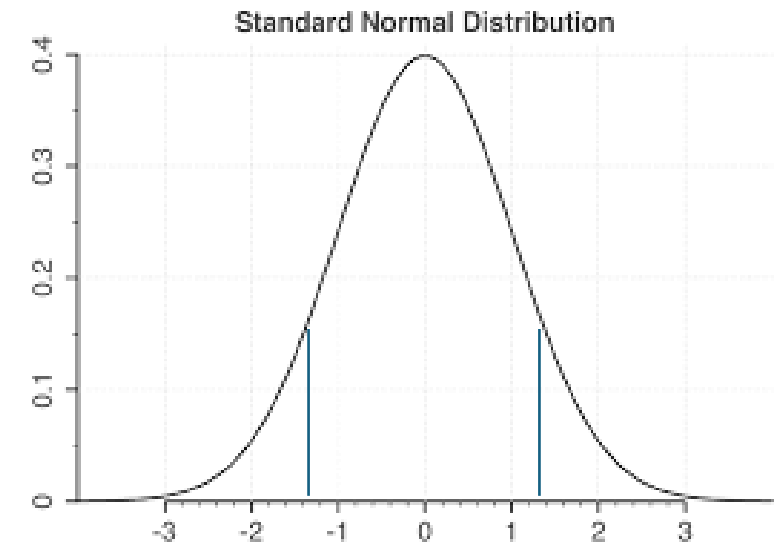
$r_2$  negative

$r_4$  positive

Prediction interval

# Prediction interval and distributional forecasts

- Recall
  - $y_{T+h|T}$  is a random variable
  - $\hat{y}_{T+h|T} = E[y_{T+h|T}]$ . The **point forecast** is the mean of the random variable.
- Assume residual  $e_{T+h} \sim N(0, \hat{\sigma}_h^2)$ 
$$y_{T+h|T} = \hat{y}_{T+h|T} + e_{T+h}$$
- Then, the **distribution** of the forecast is
$$y_{T+h|T} \sim N(\hat{y}_{T+h|T}, \hat{\sigma}_h^2)$$
- Here,  $\hat{\sigma}_h$  is the estimate of the standard deviation of the  $h$ -step forecast distribution.
- The 95% **prediction interval (PI)** for the the  $h$ -step forecast is
$$\hat{y}_{T+h|T} \pm 1.96\hat{\sigma}_h$$
- More generally, a prediction interval can be written as
$$\hat{y}_{T+h|T} \pm c\hat{\sigma}_h$$
where the **multiplier**  $c$  depends on the coverage probability.



| Percentag<br>e | Multiplier               |
|----------------|--------------------------|
| 80             | $\Phi^{-1}(0.90) = 1.28$ |
| 90             | 1.64                     |
| 95             | 1.96                     |
| 99             | 2.58                     |

# Example: Prediction interval

## Example: Quarterly demand

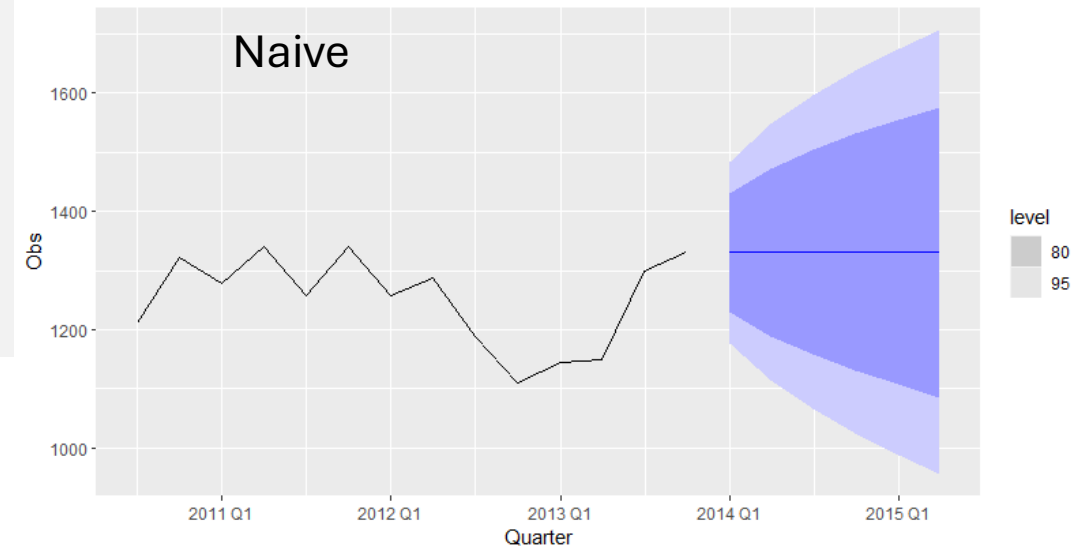
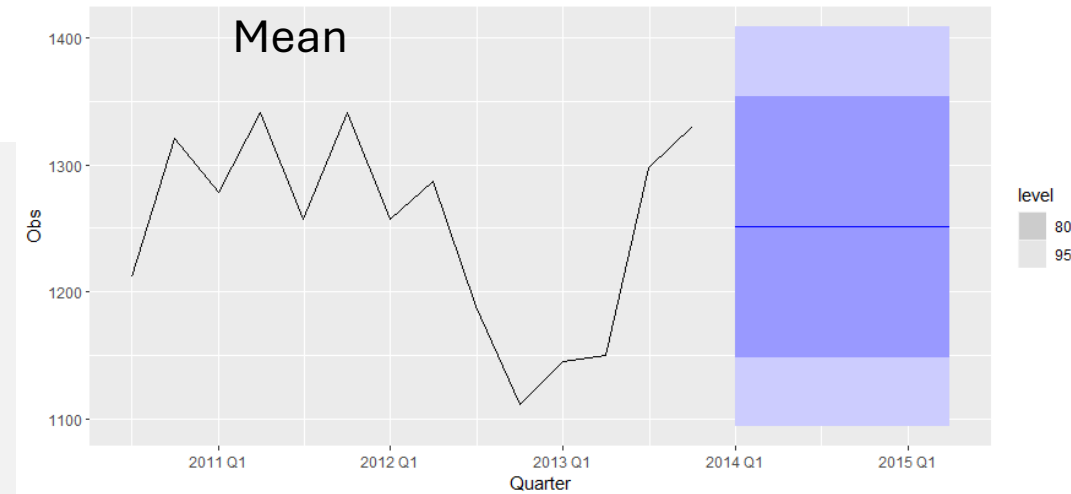
```
y <- tsibble(
  Quarter = yearquarter("2010 Q3") + 0:13,
  Obs = c(1212, 1321, 1278, 1341, 1257, 1341, 1257, 1287,
          1189, 1111, 1145, 1150, 1298, 1331),
  index = Quarter
)
```

```
y %>%
  model(MEAN(Obs)) %>%
  forecast(h=6) %>%
  autoplot(y)
```

```
y %>%
  model(NAIVE(Obs)) %>%
  forecast(h=6) %>%
  autoplot(y, level=c(80, 95))
```

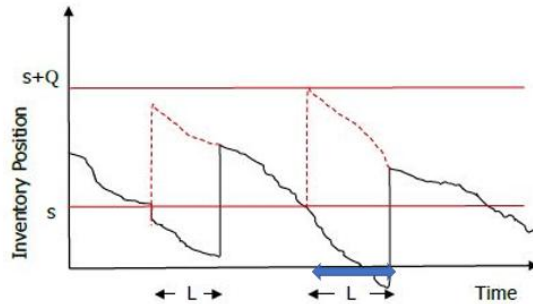
| Benchmark method | $h$ -step forecast standard deviation          |
|------------------|------------------------------------------------|
| Mean             | $\hat{\sigma}_h = \hat{\sigma} \sqrt{1 + 1/T}$ |
| Naïve            | $\hat{\sigma}_h = \hat{\sigma} \sqrt{h}$       |

- Forecast horizon = 2 years
- Plot 90% prediction interval from ETS model



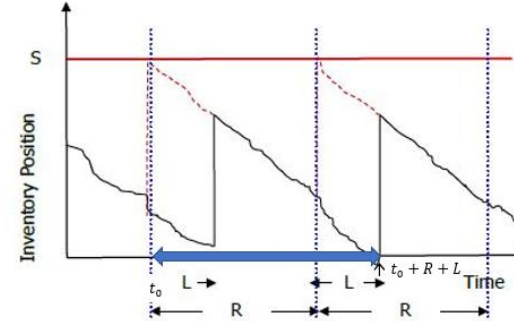
# Fundamentals of demand and supply planning\*

Continuous (s,Q) System  
 $\tau = \text{lead time (L)}$



Periodic (R,S) System

$\tau = \text{lead time plus review (L+R)}$ .

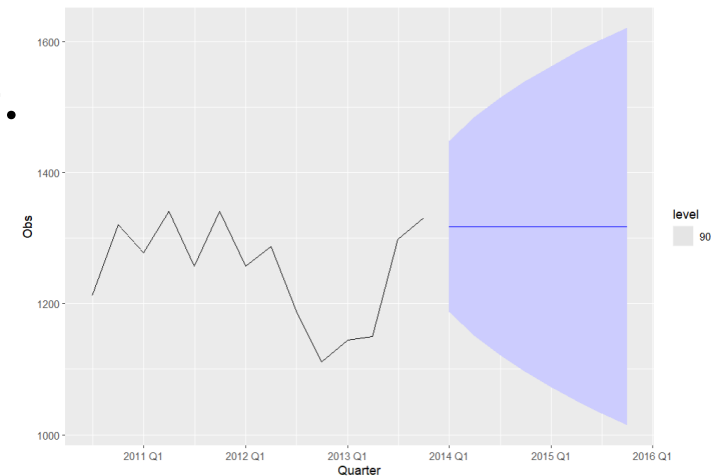


The lead time depends not only on how much time you would need to produce but also the lead time of your suppliers.

- Forecast horizon  $h$  depends on the **protection time  $\tau$  (risk period)**

$$h \geq \tau$$

- If your protection time is two months, you need to make a forecast at least two month ahead, because your production plan is locked in for those periods during which no change can be made.
- **Safety stock is affected by the forecast horizon.**
  - The longer the forecast horizon, the more uncertain will be the forecasts.
  - Uncertainty leads to more safety stocks.





# Prediction interval (PI)

- Let  $\sigma_h^2$  be the ***h*-step forecast variance**.

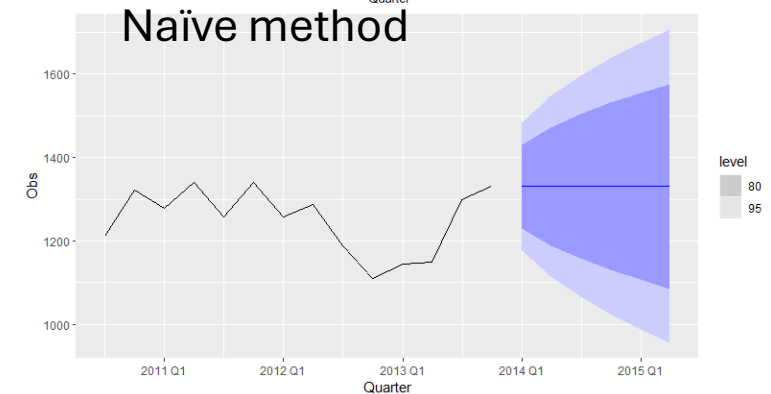
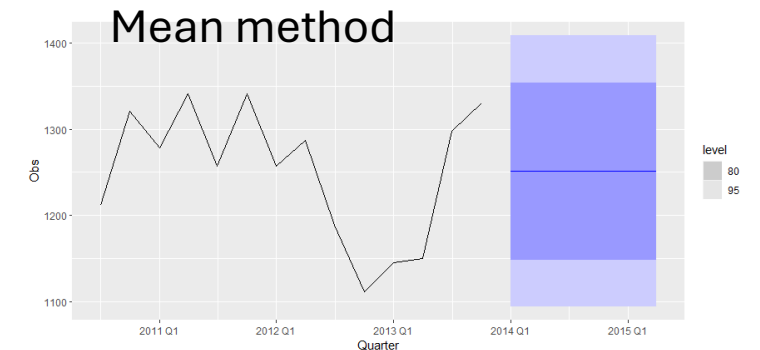
| Benchmark method | <i>h</i> -step forecast standard deviation                                  |
|------------------|-----------------------------------------------------------------------------|
| Mean             | $\hat{\sigma}_h = \hat{\sigma} \sqrt{1 + 1/T}$ where T is # of observations |
| Naive            | $\hat{\sigma}_h = \hat{\sigma} \sqrt{h}$                                    |

- Let  $z_q = \Phi^{-1}(q)$  denote the  $q$ th quantile of a standard normal distribution.

- The  $100(1 - \alpha)\%$  prediction interval is

$$\hat{y}_{T+h|T} \pm \Phi^{-1}(\alpha/2) \sigma_h$$

|                    |                           |
|--------------------|---------------------------|
|                    | Wider prediction interval |
| Forecast horizon   | Shorter or longer?        |
| Confidence level   | Higher or lower?          |
| Forecasting method | Mean or Naïve?            |



| Percentage | Multiplier |
|------------|------------|
| 50         | 0.67       |
| 55         | 0.76       |
| 60         | 0.84       |
| 65         | 0.93       |
| 70         | 1.04       |
| 75         | 1.15       |
| 80         | 1.28       |
| 85         | 1.44       |
| 90         | 1.64       |
| 95         | 1.96       |
| 96         | 2.05       |
| 97         | 2.17       |
| 98         | 2.33       |
| 99         | 2.58       |

# Standard deviation of the $h$ -step forecast $\hat{\sigma}_h^*$

- The one-step forecast

- When  $h = 1$ ,  $\hat{\sigma}_h := \hat{\sigma}$  can be estimated from residuals

$$\hat{\sigma} = \sqrt{\frac{1}{T-K} \sum_{t=1}^T e_t^2},$$

- $K$  is the number of parameters estimated in the forecasting method.
    - $K = 0$  for naïve and seasonal naïve method
    - $K = 1$  for average method and drift method

- The  $h$ -step forecast

Table 5.2: Multi-step forecast standard deviation for the four benchmark methods, where  $\sigma$  is the residual standard deviation,  $m$  is the seasonal period, and  $k$  is the integer part of  $(h-1)/m$  (i.e., the number of complete years in the forecast period prior to time  $T+h$ ).

| Benchmark method | $h$ -step forecast standard deviation             |
|------------------|---------------------------------------------------|
| Mean             | $\hat{\sigma}_h = \hat{\sigma} \sqrt{1 + 1/T}$    |
| Naïve            | $\hat{\sigma}_h = \hat{\sigma} \sqrt{h}$          |
| Seasonal naïve   | $\hat{\sigma}_h = \hat{\sigma} \sqrt{k+1}$        |
| Drift            | $\hat{\sigma}_h = \hat{\sigma} \sqrt{h(1 + h/T)}$ |

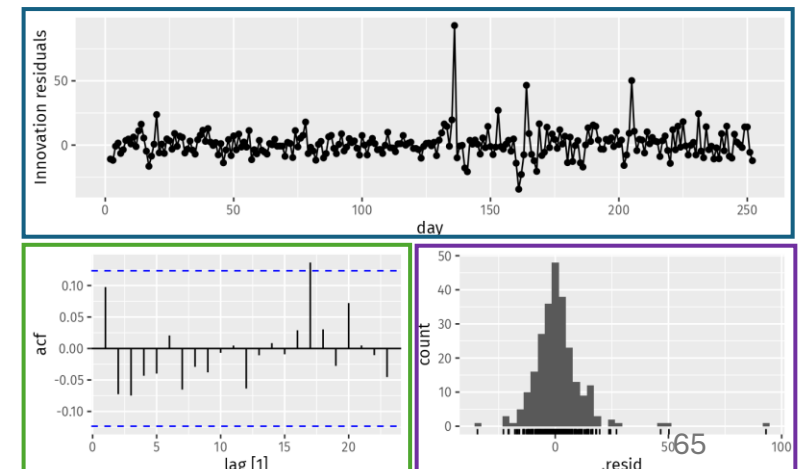
- Note that when  $h = 1$  and  $T \rightarrow \infty$ , these all give the same approximate value  $\hat{\sigma}$

# Residual diagnostics: Normality assumption

- What does this **code** do?
- From **time plot of residuals**:
  - The mean of the residuals is close to zero
  - There is no significant correlation in the residuals series. This can also be seen from **ACF**.
  - The variation of the residuals stays much the same across the historical data, apart from the one outlier, and therefore the residual variance can be treated as constant
- From the **histogram**, residuals may not be normal — the right tail seems a little too long, even when we ignore the outlier.
- Consequently, forecasts from this method will probably be quite good, but prediction intervals that are computed assuming a normal distribution may be inaccurate.



```
#####  
# Example 2 google  
# Re-index based on trading days  
google_stock <- gafa_stock |>  
  filter(Symbol == "GOOG", year(Date) >= 2015) |>  
  mutate(day = row_number()) |>  
  update_tsibble(index = day, regular = TRUE)  
# Filter the year of interest  
google_2015 <- google_stock |> filter(year(Date) == 2015)  
# Fit the models  
google_fit <- google_2015 |>  
  model(  
    Mean = MEAN(Close),  
    'Naive' = NAIVE(Close),  
    Drift = NAIVE(Close ~ drift())  
  )  
# Produce forecasts for the trading days in January 2016  
google_jan_2016 <- google_stock |>  
  filter(yearmonth(Date) == yearmonth("2016 Jan"))  
google_fc <- google_fit |>  
  forecast(new_data = google_jan_2016)  
# Plot the forecasts  
google_fc |>  
  autoplot(google_2015, level = NULL) +  
  autolayer(google_jan_2016, Close, colour = "black") +  
  labs(y = "$US",  
       title = "Google daily closing stock prices",  
       subtitle = "(Jan 2015 - Jan 2016)") +  
  guides(colour = guide_legend(title = "Forecast"))  
  
# residual diagnostics  
google_2015 |>  
  model(NAIVE(Close)) |>  
  gg_tsresiduals()
```



# Prediction intervals from bootstrapped residuals\*

When a normal distribution for the residuals is an unreasonable assumption, one alternative is to use bootstrapping, which only assumes that the residuals are uncorrelated with constant variance.



Figure 5.15: Five simulated future sample paths of the Google closing stock price based on a naïve method with bootstrapped residuals.